How Many Demonstrations Is a Task Description Worth? Specificity and Reliability in In-Context Learning

by

Trung Quan Nguyen

A thesis submitted in partial fulfillment of the requirements for the degree of

Master of Science

in

Statistics

Department of Mathematical and Statistical Sciences
University of Alberta

© Trung Quan Nguyen, 2026

Abstract

Prompts for in-context learning (ICL) describe a task, demonstrate it through examples, or both. Yet how descriptions work, especially in relation to demonstrations, remains unclear. Through the Bayesian view of ICL, we model descriptions as informative priors, characterized by their specificity and reliability. We adapt the effective sample size (ESS) from Bayesian statistics to quantify a description's predictive value as an equivalent number of demonstrations. Our theoretical results on in-context linear regression tasks show that a reliable description's ESS grows with its specificity and with the task's dimension. However, the ESS falls sharply if the description may be irrelevant: in a 16-dimensional task with no demonstrations, a one-in-ten chance of irrelevance cuts a precise description's ESS from 117 to 13.5. We then test whether Transformer networks and pretrained language models, including frontier models, attain these values. Meta-trained Transformers come close to the Bayes-optimal ESS and adapt to unreliable descriptions when trained on them. Pretrained language models, however, place too little weight on precise descriptions when answering without reasoning, and closely match the Bayesian reference only when reasoning is enabled. Taken together, these results extend the Bayesian analysis of ICL from demonstrations to descriptions.

Preface

This thesis is an original work by Trung Quan Nguyen, carried out under the supervision of Professor Linglong Kong and Professor Bei Jiang. The author formulated the research questions, designed and carried out the analysis and experiments, and wrote the text; the supervisors provided guidance and feedback throughout.

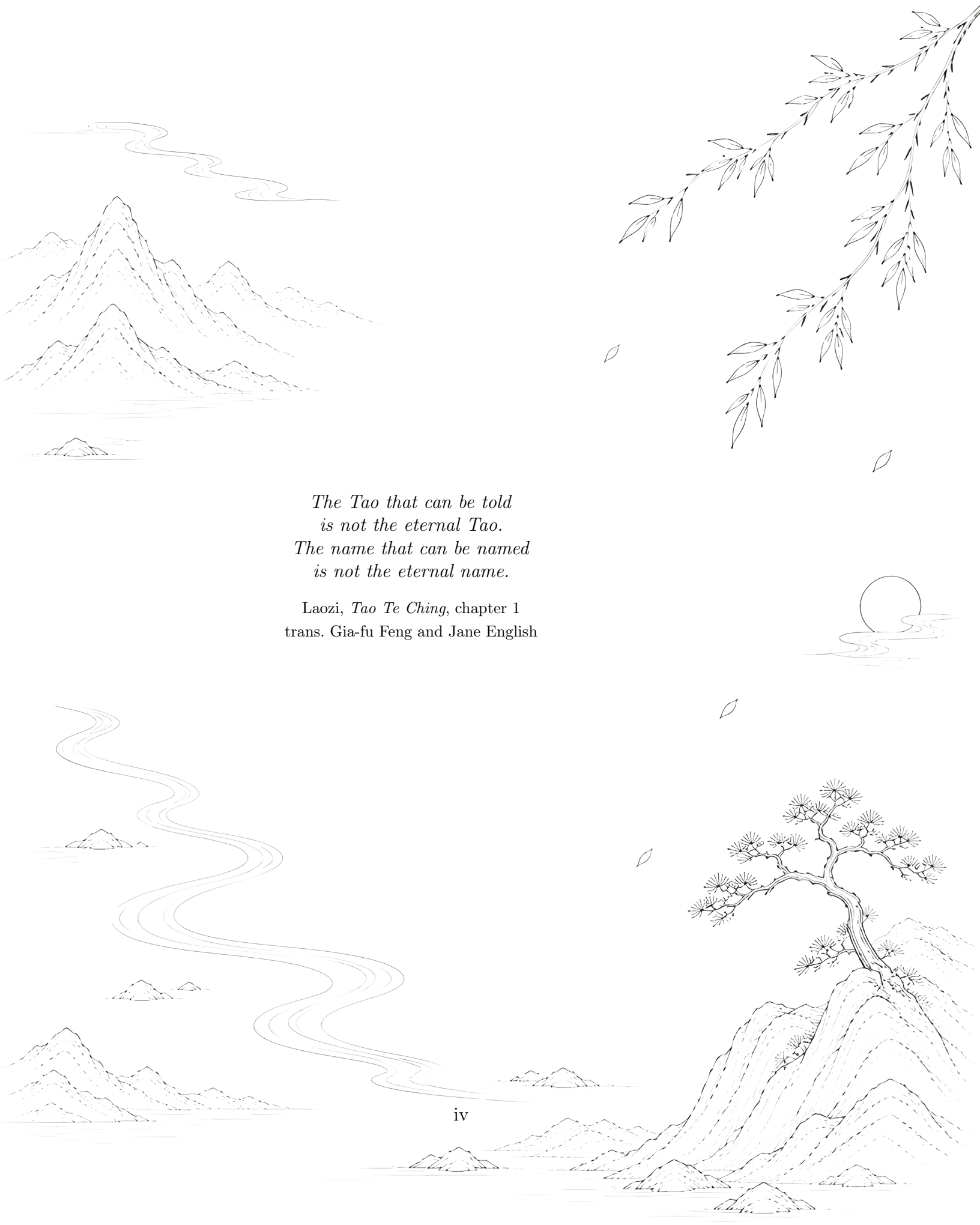
The thesis is based on the manuscript "How Many Demonstrations Is a Task Description Worth? Specificity and Reliability in In-Context Learning" by Bruce Nguyen, the name under which the author writes, prepared in the format of the International Conference on Machine Learning. Chapters 1 and 3 to 8 and Appendix A follow that manuscript's text, with its appendices placed in the chapters they support and some added detail; Sections 1.2 and 8.2 are new. Chapter 2 is new, except that it incorporates the manuscript's review of related work, placed in the sections whose topics it concerns. The manuscript has not yet been published.

The research involved no human participants, animals, or personal data, so no research ethics approval was required.

Beyond studying large language models in Chapter 7, the author used tools built on them, Anthropic's Claude Code and OpenAI's Codex, to help write and run experiment code, check derivations, and draft and edit text. The author designed the study, reviewed all code, results, and text, and takes responsibility for them.

The author's salary and the Claude Code and Codex accounts used in this research were paid for by Professor Andrew McGee of the Department of Economics, from funding for his own research.

The author founded Decaude, and this research is part of its research program; the manuscript lists the author's affiliations as Decaude and the University of Alberta.



The Tao that can be told
is not the eternal Tao.
The name that can be named
is not the eternal name.

Laozi, *Tao Te Ching*, chapter 1
trans. Gia-fu Feng and Jane English

Acknowledgements

I am deeply grateful to my family for their unconditional love, which carried me through these years.

I thank my supervisors, Professor Linglong Kong and Professor Bei Jiang, for their guidance and support throughout my program.

I am grateful to Professor Andrew McGee of the Department of Economics for his trust, patience, and generosity.

Finally, I thank God for granting me the strength and wisdom to overcome a life-altering illness during my studies and to complete this work.

Table of Contents

Abstract ii

Preface iii

Quotation iv

Acknowledgements v

List of Tables viii

List of Figures ix

1 Introduction 1

1.1 Main Findings . . . 2

1.2 Outline . . . 3

2 Background 5

2.1 In-Context Learning . . . 5

2.2 In-Context Learning as Bayesian Inference . . . 6

2.3 In-Context Linear Regression . . . 7

2.4 Squared Error . . . 8

2.5 The Effective Sample Size of a Prior . . . 10

2.6 Robust Mixture Priors . . . 12

2.7 Descriptions in Theories of ICL . . . 13

2.8 Descriptions and Priors in Pretrained Language Models . . . 14

3 Framework 16

4 The Value of a Reliable Description **19**
4.1 Complete Descriptions . . . 19
4.2 Partial Descriptions . . . 23

5 The Cost of Unreliability **26**
5.1 Unreliable Descriptions . . . 26
5.2 Assuming Relevance Under Misspecification . . . 32

6 Meta-Trained Transformers **33**
6.1 Training and Evaluation . . . 34
6.2 Matched Training and Test Reliability . . . 35
6.3 Reliability Shifts . . . 38

7 Pretrained Language Models **40**
7.1 Task and Evaluation . . . 40
7.2 Reasoning and Stated Precision . . . 44
7.3 Stated Reliability . . . 45
7.4 A Semantic Task . . . 48
7.5 Additional Results and Evaluation Details . . . 49
7.5.1 Prompts, Tasks, and Model Readouts . . . 49
7.5.2 ESS Estimation and the Bayesian Reference . . . 52
7.5.3 Results of the Full Run . . . 54
7.5.4 Irrelevant Descriptions and Wording . . . 56
7.5.5 Reasoning Runs and Execution Settings . . . 57
7.5.6 Prediction Rules and Relevance Inference . . . 60
7.5.7 Wording with Reasoning Enabled . . . 65
7.5.8 Details of the Semantic Task . . . 66

8 Discussion **70**
8.1 Scope and Limitations . . . 71
8.2 Future Work . . . 72

Bibliography **74**

A Language-Model Results at Every Number of Demonstrations **80**

List of Tables

4.1 Demonstrations can increase the value of a precise partial description 24

6.1 Transformer ESS and fitted description weights under matched and shifted reliability 37

7.1 Evaluated model configurations 41
7.2 Unreliable descriptions with the stated chance of irrelevance 46
7.3 Description ESS and weights in the full run without reasoning 50
7.4 Sensitivity of summary ESS to baseline-curve estimation 55
7.5 Description weights under four wordings without reasoning 57
7.6 Description ESS across reasoning configurations 61
7.7 Description weights under stated unreliability 64
7.8 Stating the baseline prior changes description weights 66
7.9 ESS of policy notes in the semantic task 69

A.1 ESS by description condition and demonstration count: distribution readout 81
A.2 ESS by description condition and demonstration count: written readout 83
A.3 Description weights by demonstration count: distribution readout . . 84

List of Figures

1.1 Measuring a task description in demonstrations 4

5.1 Without demonstrations, reliability limits the benefit of specificity . . 28

6.1 Meta-trained Transformers approach Bayes-optimal ESS and reflect training reliability 35
6.2 Transformer ESS as a function of demonstration count 38

7.1 Language-model ESS under stated precision and reliability 43
7.2 Learning curves used to estimate language-model ESS 54

Chapter 1

Introduction

Large language models (LLMs) are capable of in-context learning (ICL): given a prompt that specifies a task through a natural-language description, demonstrations, or both, they can predict the output for a new query input (Brown et al., 2020). In practice, prompts consist mostly of system instructions, user instructions, skill files, and earlier conversation turns, which typically describe the task rather than demonstrate it (Mao et al., 2025; Patsakis et al., 2026). The balance shifts further toward description as LLMs carry out increasingly long-horizon tasks (Kwa et al., 2025), for which demonstrations are costly to provide.

Theories of ICL, however, have mostly focused on prompts consisting of demonstrations alone (Xie et al., 2022; Garg et al., 2022; Akyürek et al., 2023). The few that include descriptions study only specific types, such as a descriptor of the input distribution (Huang & Ge, 2025), a set of candidate hypotheses (Lin et al., 2025), or related datasets supplied as context (Zhu et al., 2026); none offers a systematic treatment of descriptions. We therefore ask: how can we characterize natural-language descriptions in general, so as to understand and analyze them formally? And how well do learners, whether meta-trained Transformers or pretrained language models, use descriptions in practice?

We study the first question through the Bayesian view of ICL, in which a model infers a latent task from the demonstrations (Xie et al., 2022); a description can then be modeled as a prior over tasks. Descriptions differ in how much they say about a task: "write this report as an accountant" leaves most of the task open, whereas a step-by-step workflow leaves little. They also differ in whether they are right: an outdated instruction, for example, is simply wrong. We formalize these two properties

in d-dimensional Gaussian linear regression, a standard controlled setting for ICL (Garg et al., 2022; Panwar et al., 2024). Without a description, the coefficients follow a Gaussian baseline prior with variance a_0. With probability p, a description is relevant, and the coefficients are Gaussian around its center with precision c; otherwise, the description is wrong, and we model it as independent of the task. We thus characterize a description by two quantities: its *specificity*, set by c, and its *reliability*, the prior probability p that it is relevant (Figure 1.1).

To compare descriptions with demonstrations, we adapt measures that express a prior's impact, or a predictor's performance, as a number of observations (Wiesenfarth & Calderazzo, 2020; Reimherr et al., 2021; Le Scao & Rush, 2021; Selby et al., 2025). A description's *predictive effective sample size* (ESS) is the number of additional demonstrations that a learner needs, without the description, to reach the same expected error. We compute ESS for the Bayes-optimal predictor and for trained models. The first gives the information that a description carries under the statistical model; the second gives the benefit that a learner realizes. For the second question, we test how close trained learners come to the Bayes-optimal ESS.

1.1 Main Findings

Our main findings are as follows:

- **Precision and dimension determine the value of reliable descriptions.** With precision c fixed at a positive integer, zero-demonstration ESS approaches $c + d + 1$ as the baseline prior becomes diffuse (Chapter 4).
- **Uncertain relevance limits zero-demonstration ESS.** Any fixed positive probability of irrelevance imposes a finite ceiling as precision grows; with demonstrations, a learner can also infer whether the description is relevant (Chapter 5).
- **For Transformers, description use depends on training; for language models, on reasoning.** Transformers trained from scratch on the Gaussian model must learn a description's precision and reliability from their training data. They approach the Bayes-optimal ESS, and the weight they give a description follows the reliability of that data. Pretrained language models are instead tested on numerical mean-estimation tasks whose prompts state both properties; for most models tested, enabling reasoning brings their use of the stated precision close to the Bayesian

reference (Chapters 6–7).

1.2 Outline

Chapter 2 reviews in-context learning and its Bayesian interpretation, in-context linear regression and its squared error, the effective sample size of a prior, and robust mixture priors, together with related work. Chapter 3 presents the framework: tasks and demonstrations, descriptions as uncertain task information, prediction error, and predictive ESS. Chapter 4 derives the value of a reliable description, with its proofs and the analysis of partial descriptions. Chapter 5 derives the cost of unreliability, with its proofs and the analysis of the Bayesian predictor assuming relevance under misspecification. Chapter 6 tests meta-trained Transformers, with their training details and further networks. Chapter 7 tests pretrained language models on numerical prediction tasks with explicit probabilistic descriptions, with the evaluation details and further experiments on stated reliability and on a semantic classification task. Chapter 8 discusses the findings and their limitations, and outlines future work. Appendix A reports the language-model tables at every evaluated demonstration count.

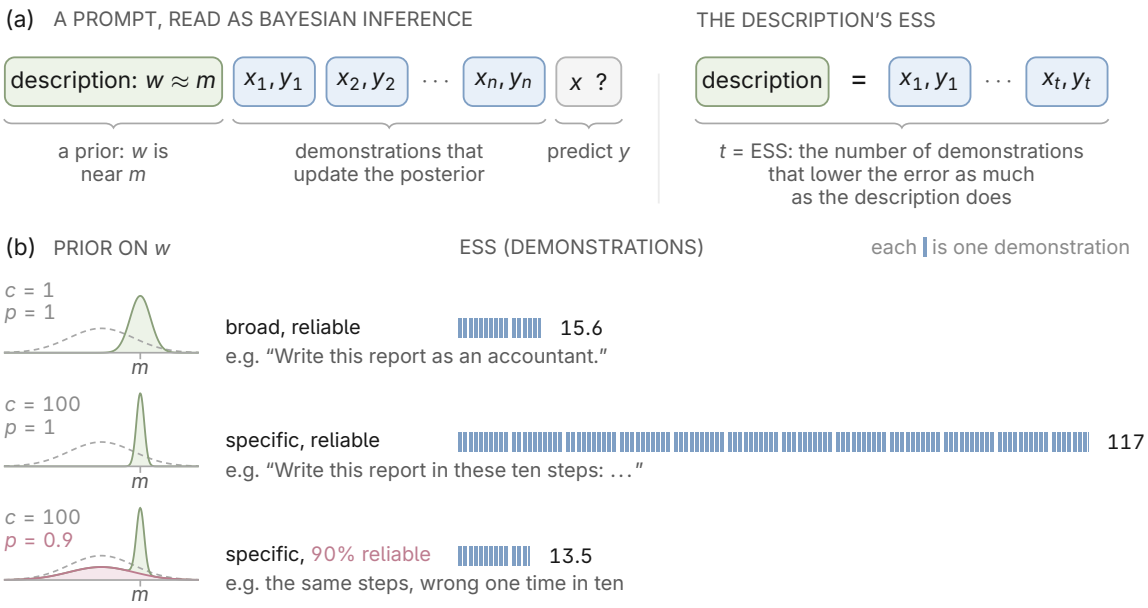


Figure 1.1: **Measuring a task description in demonstrations.** (a) A description provides information about a latent task w, and demonstrations supply additional evidence. A description's predictive effective sample size (ESS) is the number of additional demonstrations that a learner would need, in place of the description, to reach the same error. The schematic on the right illustrates $n = 0$. (b) Description-conditioned task distributions and their Bayes-optimal ESS at $n = 0$, $d = 16$, and $a_0 = 10$. Dashed curves show the baseline prior, and pink the irrelevant component. Specificity is set by the precision c of the relevant component; reliability p is that component's prior probability. The example instructions illustrate each kind of description; the ESS values are those of the Gaussian model. Increasing specificity greatly increases ESS when $p = 1$, whereas a 10% probability of irrelevance sharply limits this benefit.

Chapter 2

Background

In this chapter, we review the ideas that the rest of the thesis builds on. Sections 2.1 to 2.3 introduce in-context learning, its interpretation as Bayesian inference, and in-context linear regression, a setting in which that interpretation can be computed exactly. Section 2.4 defines the squared error by which we compare learners and gives the error of the Bayesian predictor. Sections 2.5 and 2.6 present two tools from Bayesian statistics, the effective sample size of a prior and the robust mixture prior. We later apply both to descriptions, and these sections relate our use of them to other work. Finally, Section 2.7 reviews theories of ICL that include descriptions, and Section 2.8 reviews studies of how pretrained language models use descriptions and stated probabilities.

2.1 In-Context Learning

A language model takes a sequence of tokens, the prompt, and predicts what comes next. Brown et al. (2020) showed that large language models can perform a task specified in the prompt, without any update to their weights. This ability is called in-context learning (ICL). Brown et al. (2020) distinguished three kinds of prompts. A zero-shot prompt gives only a natural-language description of the task. One-shot and few-shot prompts add one or several *demonstrations*, solved examples of the task, before the query that the model must answer. We follow their terminology: a demonstration is one input together with its label, and a description gives information about the task other than through demonstrations, whether as a statement about

the task or as a command. Weller et al. (2020) likewise formalize learning from task descriptions by distinguishing the latent task, its natural-language description, and individual input–output examples. Chapter 3 gives the model of a description that we analyze. In our experiments, the meta-trained Transformers receive a description as a numeric token, and the language models receive it as a probabilistic statement or a policy note. A prompt can contain both a description and demonstrations, and our question is how much the description adds to the demonstrations.

How models use the demonstrations is itself an active question. Pretrained models can perform well even when the labels of the demonstrations are replaced by random ones (Min et al., 2022), which suggests that they use demonstrations partly to recognize a task they already know. Larger models, however, follow flipped in-context labels more readily (Wei et al., 2023). Lin & Lee (2024) explain both behaviors, retrieving a known task and learning a new one, within a single Bayesian model. A description raises the same question from the other side. Does a model treat it as information about the task, to be weighed against the demonstrations, or does it use it in some other way?

2.2 In-Context Learning as Bayesian Inference

Theoretical work studies ICL as Bayesian inference over latent tasks (Xie et al., 2022; Panwar et al., 2024), as a learning algorithm implemented in a forward pass (Akyürek et al., 2023; von Oswald et al., 2023), and as meta-learning across tasks (Garg et al., 2022; Müller et al., 2022). Meta-trained predictors can approximate Bayesian inference under suitable training conditions (Genewein et al., 2025), and the learning algorithms that Transformers implement in context approach the Bayes-optimal estimator as the networks grow (Akyürek et al., 2023). Agreement with the Bayesian predictor can depend on training-task diversity and on the function class (Raventós et al., 2023; Panwar et al., 2024), and language-model ICL can violate Bayesian consistency (Falck et al., 2024). We adopt the Bayesian view, because it extends naturally from demonstrations to descriptions (Chapter 3).

In the Bayesian view, as Xie et al. (2022) proposed it, pretraining text is generated from latent concepts. A model that predicts such text well must then infer the concept behind each document, and a prompt of demonstrations is evidence about the concept behind the prompt. The model's prior over concepts is acquired in pretraining, so

even a prompt without demonstrations draws on it, and a description refines it before any demonstration is seen. Formally, a task is a latent parameter w, and a prompt contains n demonstrations $D_n = \{(x_i, y_i)\}_{i=1}^n$ and a query x. The ideal learner predicts the query's label with the posterior predictive distribution

$$f(y \mid x, D_n) = \int f(y \mid x, w)\, f(w \mid D_n)\, dw. \tag{2.1}$$

This view becomes testable when the prior over tasks is known. Consider a Transformer trained on prompts whose tasks are sampled from a known prior. Training on such prompts is called meta-training: the network is trained on many tasks and must infer from the context which task a prompt comes from. The posterior predictive minimizes the expected log loss over the prior, and this expected log loss is exactly the training objective. A well-trained network therefore approximates the posterior predictive (Müller et al., 2022), whose mean is the Bayes-optimal prediction under squared error. Section 2.4 shows both properties of the posterior predictive: that it minimizes expected log loss, and that its mean minimizes squared error. For regression with a Gaussian task prior, Transformers meta-trained on enough distinct tasks behave like the Bayesian predictor, which is ridge regression (Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024); Section 2.3 describes these experiments. The Bayesian predictor is then a benchmark against which a network can be measured, and Chapter 6 uses it in this way.

2.3 In-Context Linear Regression

Garg et al. (2022) introduced in-context linear regression as a controlled test of ICL. Each prompt has its own task vector $w \sim \mathcal{N}(0, I_d)$, its inputs are drawn as $x_i \sim \mathcal{N}(0, I_d)$, and its labels are $y_i = w^\top x_i$. Garg et al. (2022) found that a Transformer trained from scratch on such prompts predicts the query's label about as well as least squares. Later work added label noise, $y_i = w^\top x_i + \epsilon_i$ (Akyürek et al., 2023; Raventós et al., 2023). Raventós et al. (2023) also varied the number of distinct tasks seen in training. With few tasks, the network behaves like the Bayesian predictor for a prior over those tasks alone. With enough tasks, it behaves like ridge regression, the Bayesian predictor for the Gaussian prior.

With Gaussian noise and a Gaussian prior, Bayes-optimal prediction has a closed form (Gelman et al., 2013, Section 14.8). We measure labels in units of their noise, so

that $y_i \sim \mathcal{N}(w^\top x_i, 1)$. Let the prior be $w \sim \mathcal{N}(m, \Lambda^{-1})$, with center m and precision Λ. Let $X_n \in \mathbb{R}^{n \times d}$ contain the demonstration inputs as rows, and write $W_n = X_n^\top X_n$. The posterior over w is Gaussian, with covariance and mean

$$\Sigma = \big(\Lambda + W_n\big)^{-1}, \qquad \mu = \Sigma\big(\Lambda m + X_n^\top y_{1:n}\big), \tag{2.2}$$

and the posterior predictive for the query is

$$y \mid x, D_n \sim \mathcal{N}\big(\mu^\top x,\ 1 + x^\top \Sigma x\big). \tag{2.3}$$

The predictive variance has two parts. The first, 1, is the label noise. The second, $x^\top \Sigma x$, is the remaining uncertainty about the task in the direction of the query. Under the baseline prior $\mathcal{N}(0, a_0 I_d)$, the posterior mean $\mu = (W_n + a_0^{-1} I_d)^{-1} X_n^\top y_{1:n}$ is the ridge regression estimate with penalty a_0^{-1}. Equation (2.2) also shows that a prior acts like additional data. Each demonstration adds $x_i x_i^\top$ to the posterior precision, and the prior adds Λ. Chapter 3 uses this fact to give a description a prior of its own, and in Chapter 4 a reliable description replaces the baseline precision a_0^{-1} with c.

2.4 Squared Error

A predictor maps the prompt to a prediction $\hat{y}$ of the query's label. We score it by squared error. The label is $y = w^\top x + \epsilon$, where the noise ϵ has mean 0 and variance 1 and is independent of the prompt and the task. The expected squared error therefore splits into two parts:

$$\mathbb{E}\big[(\hat{y} - y)^2\big] = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big] + 1. \tag{2.4}$$

The second part is the noise variance, which is the same for every predictor. Following Garg et al. (2022), we therefore score a predictor by the first part alone. We call it the predictor's *prediction error*, or *risk*:

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big].$$

It is the expected squared distance from the noise-free label $w^\top x$. A predictor that knows the task predicts $w^\top x$ and has zero error.

Given the prompt, the task is uncertain. For any prediction $\hat{y}$ computed from the prompt,

$$\mathbb{E}\big[(\hat{y} - w^\top x)^2 \mid x, D_n\big] = \big(\hat{y} - \mathbb{E}[w^\top x \mid x, D_n]\big)^2 + \mathrm{Var}(w^\top x \mid x, D_n).$$

The first term vanishes only when $\hat{y}$ is the posterior mean of $w^\top x$. When tasks are drawn from the prior, the posterior mean therefore has the smallest error of any predictor. We call it the Bayesian predictor and its error Bayes-optimal. It is also the mean of the posterior predictive (2.1). Its error is the expected posterior variance of $w^\top x$. In linear regression, the posterior is Gaussian with covariance Σ (Equation (2.2)), so the posterior variance of $w^\top x$ is $x^\top \Sigma x$, the part of the predictive variance (2.3) that comes from uncertainty about the task. The query input is independent of the demonstrations, with $\mathbb{E}[xx^\top] = I_d$, so the error of the Bayesian predictor is

$$R = \mathbb{E}\left[x^\top \Sigma x\right] = \mathbb{E}\,\mathrm{tr}\,\Sigma, \tag{2.5}$$

the expected trace of the posterior covariance. The error is zero when the posterior has concentrated on the task, and it grows with the uncertainty about the task that remains.

Two standard facts give this trace in the cases that later chapters need. Chapter 3 writes $R^{\mathrm{demo}}(n)$ for the error with n demonstrations and no description. With fewer demonstrations than dimensions, $n < d$, the demonstrations carry no information about $d - n$ directions, so under the baseline prior $\mathcal{N}(0, a_0 I_d)$ we have $R^{\mathrm{demo}}(n) \geq (d-n)\, a_0$. In the limit $a_0 \to \infty$ of a flat prior, and with $n \geq d + 2$ demonstrations, the posterior covariance is W_n^{-1}, the inverse of a Wishart matrix. The error is then $\mathbb{E}\,\mathrm{tr}\, W_n^{-1} = d/(n - d - 1)$ (Muirhead, 1982). Chapter 4 builds its limits for a reliable description (Proposition 4.1) from these two facts.

A predictor that gives a density $g(y \mid x, D_n)$ rather than a number can instead be scored by its log loss, $-\log g(y \mid x, D_n)$. The expected log loss is smallest for the true conditional density of y given the prompt (Cover & Thomas, 2006), which is the posterior predictive when tasks are drawn from the prior. Meta-trained networks are often trained with this loss (Section 2.2), and so are the networks of Chapter 6. The mean of the posterior predictive is the posterior mean, so a network that minimizes log loss and outputs a predictive mean can be scored by squared error. We use squared error throughout for two reasons. First, it requires only a point prediction, so it applies to the number a language model writes (Chapter 7). Second, the Bayes-optimal error (2.5) has the exact forms above, so the theory in Chapters 4 and 5 can work with exact errors.

Squared error also splits exactly over the components of a mixture. Suppose that a latent indicator Z selects the component that generates the task. By the law of total

variance, the posterior variance is the sum of two parts:

$$\begin{aligned}\operatorname{Var}(w^\top x \mid x, D_n) = \mathbb{E}\big[\operatorname{Var}(w^\top x \mid x, D_n, Z) \mid x, D_n\big] \\ + \operatorname{Var}\big(\mathbb{E}[w^\top x \mid x, D_n, Z] \mid x, D_n\big).\end{aligned} \tag{2.6}$$

The first part is the average variance within the components. The second is the variance of the prediction across the components. Chapter 5 applies this decomposition to a description that may be irrelevant. There, the first part gives the risk of the Bayes-optimal predictor that also observes Z, and the second is the excess Bayes risk due to unobserved relevance.

2.5 The Effective Sample Size of a Prior

Bayesian statistics measures the information in a prior by its effective sample size (ESS), the number of observations that would carry the same information. In the conjugate normal model, this number is exact. Suppose that the observations are $y_i \sim \mathcal{N}(w, \sigma^2)$ and the prior is $w \sim \mathcal{N}(m, \sigma^2/c)$. After n observations with mean $\bar{y}$, the posterior is

$$w \mid y_{1:n} \sim \mathcal{N}\Big(\frac{c\,m + n\bar{y}}{c+n},\ \frac{\sigma^2}{c+n}\Big).$$

This is the posterior that $c+n$ observations would give under a flat prior if c of them averaged m. The prior is therefore worth c observations: c is its precision in units of the precision of one observation. Chapter 7 uses this model for its prompt-based Bayesian reference. The precision c of a description (Chapter 3) is this ESS in each direction of the task, counted in direct observations of the task with unit noise.

Beyond the conjugate case, how many observations a prior is worth depends on what is matched, and the literature offers several choices (Morita et al., 2008; Neuenschwander et al., 2020). Morita et al. (2008) match information, which they measure by the curvature of the log density. Their ESS is the number of hypothetical observations that would turn a very vague prior into a posterior carrying about the same information as the prior. Neuenschwander et al. (2020) add a consistency requirement: on average, the ESS of the posterior after N observations should equal the prior's ESS plus N. Other definitions match performance at the current sample instead. Reimherr et al. (2021) count the observations that a baseline prior would need to match a given prior's expected posterior loss at a given sample size; their ESS can

be negative when the prior and the likelihood conflict. The effective current sample size of Wiesenfarth & Calderazzo (2020) starts from the accuracy of an analysis of the current sample with the prior. It counts the observations that must be added to the sample, or removed from it, for an analysis without the prior to be equally accurate. Reznik (2026) uses predictive log loss to choose the discount of a power prior, which raises the likelihood of historical data to a power between 0 and 1.

Our aim is to compare what a description is worth to different learners: the Bayes-optimal predictor, meta-trained networks, and pretrained language models. Matching information requires an explicit prior, which only the first of these has. A network or a language model also brings prior knowledge to a task, acquired in training, but that knowledge is implicit in its weights, and only its predictions can be observed. Matching performance requires only those predictions. Our predictive ESS (Definition 3.1) therefore follows the loss-based measures of Reimherr et al. (2021) and Wiesenfarth & Calderazzo (2020): it is the number of additional demonstrations that a learner needs, without the description, to reach the prediction error that it has with the description. It differs from those measures in three ways: it evaluates population prediction error, it allows a non-Bayesian learner, whose own curve without the description serves as the baseline, and it depends on the number of demonstrations already observed. The word *predictive* names what is matched; it does not assert the predictive-consistency property of Neuenschwander et al. (2020).

Matching performance rather than information changes the answer even for the Bayes-optimal predictor. By the measure of Morita et al. (2008), a prior of precision c adds c to the posterior precision regardless of the sample size, so a description that raises the precision from a_0^{-1} to c in every direction is worth $c - a_0^{-1}$ demonstrations, regardless of how many have been observed. Prediction error behaves differently when the inputs are random. In d dimensions under a flat prior, the Bayesian predictor's error after n demonstrations is $d/(n - d - 1)$ (Section 2.4). The expected information of the demonstrations would suggest d/n; the error is larger because random inputs constrain some directions less than others. Chapter 4 shows the consequence. An ESS that matches error exceeds $c - a_0^{-1}$ by a term that grows with the dimension and vanishes as demonstrations accumulate. With few demonstrations, a reliable description is therefore worth more than its added precision; with many, its ESS approaches the value of Morita et al. (2008).

Two predictive comparisons in machine learning are especially close to ours. Le Scao

& Rush (2021) measure the horizontal separation of learning curves in prompt-based fine-tuning, averaged over a range of accuracies; we evaluate a pointwise difference in demonstration counts. Selby et al. (2025) compare priors elicited from language models with data-based predictors; their measure is the number of observations needed to match predictive loss. We instead study descriptions supplied to a learner, and how their benefit changes after demonstrations. Beyond ICL, an instruction can replace up to 100 execution examples in the fine-tuning settings studied by Cook et al. (2026). Wenliang et al. (2026) treat a prompt as conditioning for a Bayesian sequence predictor and compare an optimized prompt against a set of demonstrations: the optimized prompt can be shorter than demonstrations that have the same effect.

2.6 Robust Mixture Priors

A prior built from historical data can conflict with the current data. Evans & Moshonov (2006) check for such prior–data conflict by asking whether the observed data are surprising under the prior predictive distribution. Heavy-tailed priors accommodate such conflict (O'Hagan & Pericchi, 2012), and clinical trials that borrow historical control data protect against it with a robust mixture prior (Schmidli et al., 2014). The informative prior f_I, derived from the historical data, is mixed with a vague prior f_V:

$$f(w) = p\, f_I(w) + (1 - p)\, f_V(w).$$

The posterior is then a mixture of the two component posteriors, and the weight on the informative component is updated in proportion to p times its marginal likelihood. When the data agree with the informative component, its weight grows, and the analysis borrows strength from the historical data. When they conflict, the weight moves to the vague component, and the analysis falls back on the current data.

A description is a prior of the same kind. It may describe the task at hand, but it may also not apply to it, for example because it was written for another task. Chapter 3 therefore models a description as a robust mixture prior whose weight is its reliability p, the probability that the description is relevant. Our mixture is specified directly, rather than derived from a historical-data meta-analysis. Chapter 5 calls the updated weight the posterior relevance probability π_n and shows how reliability affects the description's ESS. Averaging the component predictions with weights π_n is Bayesian model averaging (Hoeting et al., 1999). Chapter 7 also compares language-model

predictions with *fixed-weight averaging*, which keeps the weights at p and $1-p$ instead of updating them.

In our model, an irrelevant description is independent of the task and may happen to agree with it. Irrelevance is therefore a property of how the description was generated, and it differs from the prior–data conflict above, which concerns disagreement between prior information and observed data.

2.7 Descriptions in Theories of ICL

Theories of ICL have added task information to demonstrations in several forms: a token that carries the mean of the input distribution (Huang & Ge, 2025), tokens that indicate a class of candidate hypotheses (Lin et al., 2025), and related datasets supplied as context (Zhu et al., 2026). Other theories study a prior that mixes task groups (Lin & Lee, 2024) and an instruction whose influence changes (Tong et al., 2026). These theories generally do not measure the value of a description in units of demonstrations, and they do not model the possibility that it is irrelevant. For example, Huang & Ge (2025) show that a learner can use the description token to predict better, but they do not say how many demonstrations it is worth, or what it is worth when it may be irrelevant. The work closest to our setting is that of Zhu et al. (2026), who show that a Transformer given a prefix of related datasets performs amortized hierarchical Bayesian prediction. We use explicit Bayesian predictors to quantify how such conditioning information changes the number of demonstrations needed to reach a given prediction error, and we allow that information to be irrelevant.

The Bayesian theory of Xie et al. (2022) models the prompt as demonstrations alone. We keep its view of the prompt as evidence about a latent task and add a description, which induces a distribution over tasks before the demonstrations are observed. Our networks in Chapter 6 use the prefix embedding of Huang & Ge (2025), in which the description is a single token placed before the demonstrations.

2.8 Descriptions and Priors in Pretrained Language Models

Empirical studies differ on how much descriptions help language models. Task summaries distilled from demonstrations can replace the examples in context (Honda et al., 2025), whereas other studies report weak sensitivity to instruction relevance (Webson & Pavlick, 2022) or limited benefits from repository instruction files (Gloaguen et al., 2026). When a description helps little, these studies cannot tell whether it carried little information or the model failed to use it. Our controlled experiments instead specify the joint distribution of task, description, and demonstrations, so the information a description carries is known and a model's use of it can be measured against a Bayesian reference.

Other work evaluates language-model predictions against Bayesian references. Gupta et al. (2025) compare explicitly stated coin biases with observations in this way. The learning curves of pretrained models approach a Bayesian reference on synthetic in-context function-learning tasks with a known optimum (Akata et al., 2026), and Arora et al. (2025) model in-context learning curves with a Bayesian scaling law that includes a task prior. Qiu et al. (2026) also study how training can improve probabilistic reasoning; He et al. (2025) examine Bayesian consistency during iterative reasoning on open-ended questions; and concurrent work studies how reasoning and code execution affect language models' approximation of Bayesian posteriors (Filbry et al., 2026). Text supplied alongside numbers can also change numerical predictions (Requeima et al., 2024), and models estimate a density from numbers in context like a kernel density estimator with an adaptive bandwidth (Liu et al., 2025).

Our numerical experiments state uncertainty explicitly and vary stated precision and reliability separately, with controls that hold the description center and demonstrations fixed, so that predictions can be compared with a specified Bayesian reference. Code execution, whose effect Filbry et al. (2026) study, is disabled in our main runs. The semantic task extends this comparison to ordinary-language constraints on a finite family of classification policies.

Our settings express specificity in three forms that share no numerical scale: the prior precision c in the isotropic Gaussian model; the number k of constrained directions of a partial description (Chapter 4); and, in the semantic task of Chapter 7, the number of policies that a policy note leaves compatible. ESS puts these forms on one scale,

the number of demonstrations a description is worth. To show how a model uses a description, Chapters 6 and 7 also summarize predictions by *description weights*. A description weight is the weight that a prediction places on the description's center; we either fit one across prompts or compute one for each response. In Gaussian mean estimation, the Bayesian reference weights the prior mean and the sample mean in proportion to their precisions (Neuenschwander et al., 2020). A normalized description weight can be compared with posterior component probabilities, but need not itself be a probability.

Chapter 3

Framework

This chapter applies the tools of Chapter 2 to task descriptions: a description becomes a robust mixture prior over the regression task, and its value is measured by a predictive effective sample size.

Tasks and demonstrations. We use the noisy in-context linear regression setting of Section 2.3 (Garg et al., 2022; Raventós et al., 2023). A task is a vector $w \in \mathbb{R}^d$ with baseline prior $\mathcal{N}(0, a_0 I_d)$, where $a_0 > 0$. A prompt contains n demonstrations $D_n = \{(x_i, y_i)\}_{i=1}^n$ and a query input x, with

$$x_i, x \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d), \qquad y_i = w^\top x_i + \epsilon_i, \quad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

Inputs and observation noise are independent of the task and one another. The parameter a_0 controls variation across tasks. Numerical illustrations in Chapters 4–5 use $d = 16$ and $a_0 = 10$.

Descriptions and relevance. A *description* is information about the task w that a prompt gives other than through demonstrations, such as a numeric token given to a Transformer (Chapter 6) or a probabilistic statement or policy note given in text to a language model (Chapter 7). Anything that a prompt contains whether or not it has a description, such as instructions on the answer format, is not part of the description. We model a description by a center $m \in \mathbb{R}^d$, a precision $c \geq a_0^{-1}$ that measures its *specificity*, and its *reliability* $p \in [0, 1]$. Let $r = (ca_0)^{-1} \in (0, 1]$ be the fraction of baseline variance remaining in the relevant component. With a latent

relevance indicator $Z \sim \text{Bernoulli}(p)$ independent of m, the joint model is

$$\begin{aligned} m &\sim \mathcal{N}(0, (1-r)a_0 I_d), \\ w \mid m, Z = 1 &\sim \mathcal{N}(m, r a_0 I_d), \\ w \mid m, Z = 0 &\sim \mathcal{N}(0, a_0 I_d). \end{aligned}$$

Thus

$$w \mid m \sim p\,\mathcal{N}(m, c^{-1} I_d) + (1-p)\,\mathcal{N}(0, a_0 I_d).$$

This is a robust mixture prior (Section 2.6; Schmidli et al., 2014), with weight p on the description's component. Integrating over m recovers $w \sim \mathcal{N}(0, a_0 I_d)$ under either component, so comparisons with and without descriptions concern the same task population.

A description is *relevant* when $Z = 1$ and *irrelevant* when $Z = 0$. Relevance identifies the generating component, irrespective of the realized distance between m and w. We use *reliable* for $p = 1$ and *unreliable* for $p < 1$. At $c = a_0^{-1}$, the two components coincide and the description provides no task information. The model separates uncertainty within the described region from uncertainty about whether that region applies. It does not assume that a precision and a reliability can summarize every natural-language description.

A relevant description splits the baseline variance a_0 into the part $(1-r)a_0$ that m resolves and the part ra_0 that remains. Because the observation noise has unit variance, a description of precision c carries in each direction the precision of c direct observations of w. Since $\mathbb{E}W_n = nI_d$, this is also the expected information of c demonstrations, and it is the ESS that Morita et al. (2008) assign to a normal prior of precision c (Section 2.5). Relative to the baseline prior, the description adds precision $c - a_0^{-1}$. We call a description *broad* when c is small relative to d and *specific* when c exceeds d; Chapter 4 shows that dimension dominates the ESS of a broad description, and precision dominates the ESS of a specific one. At $d = 16$ and $a_0 = 10$, a description with $c = 1$ ($r = 0.1$) is broad, and one with $c = 100$ ($r = 0.001$) is specific.

Prediction error. As in Section 2.4, we score a prediction $\hat{y}$ by its prediction error, or risk, the expected squared error against the noise-free query response:

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big]. \tag{3.1}$$

The expectation is over tasks, descriptions, demonstrations, and queries. For a fixed learner, $R^{\mathrm{desc}}(n)$ and $R^{\mathrm{demo}}(n)$ denote risk with and without a description, and $R^{\mathrm{desc}}_{p=1}(n)$ denotes the reliable case. As a function of the demonstration count, R^{demo} is the learner's *baseline curve*. To express a description's value in demonstrations, we compare these two risk curves, as the loss-based ESS of Section 2.5 does for a prior.

Definition 3.1 (Predictive effective sample size)**.** Given n demonstrations, let n^* be the smallest $t \geq 0$ satisfying $R^{\mathrm{demo}}(t) = R^{\mathrm{desc}}(n)$. When this crossing exists, the description's ESS is

$$\mathrm{ESS}_n = n^* - n.$$

Noninteger counts refer to an interpolated baseline curve.

ESS measures the horizontal displacement between a learner's learning curves with and without the description. Without a description, a learner still predicts from prior knowledge. For the Bayes-optimal predictor, this knowledge is the baseline prior; for a network or a language model, it is what the model learned in training. ESS therefore measures what the description adds to that knowledge, its value *to that learner*. An ESS of 10 means that n demonstrations plus the description match the error of $n + 10$ demonstrations alone. Because each learner is compared with its own baseline curve, a large ESS need not imply low absolute error. For a decreasing baseline curve, negative ESS indicates higher error at the same demonstration count. For the Bayes-optimal predictor under the correctly specified model, conditioning on the description cannot increase expected risk, so $R^{\mathrm{desc}}(n) \leq R^{\mathrm{demo}}(n)$ and ESS is nonnegative. Because R averages over descriptions, the ESS is a property of a class of descriptions, determined by c and p, rather than of a particular value of m.

Empirical ESS interpolates a learner's measured risk curve. The interpolation is linear in the demonstration count for the Bayes-optimal predictor, the meta-trained Transformers, and the semantic task. For the numerical language-model tasks, whose evaluation grid doubles n, it is linear in the logarithm of the count. It uses the first crossing in the evaluated range. When no crossing is observed, we report censoring at the corresponding boundary under the conventions in Section 7.5.2; an exact ESS is then unavailable.

Chapter 4

The Value of a Reliable Description

For $p = 1$, a description with fixed positive integer precision c has $\mathrm{ESS}_0 \to c + d + 1$ as $a_0 \to \infty$. Its value without demonstrations therefore depends on task dimension as well as precision. Section 4.1 derives this limit for a description that constrains every direction. It also derives the limit $c - a_0^{-1}$ that such a description's ESS approaches as demonstrations accumulate. Section 4.2 treats a partial description, which constrains only some directions and can gain value as demonstrations resolve the rest.

4.1 Complete Descriptions

The Bayes-optimal risk is the expected trace of the posterior covariance of w (Equation (2.5)). With $W_n = X_n^\top X_n$ as in Section 2.3, Gaussian conjugacy gives the Bayes-optimal risks

$$R^{\mathrm{demo}}(n) = \mathbb{E}\,\mathrm{tr}(a_0^{-1} I_d + W_n)^{-1},$$
$$R^{\mathrm{desc}}_{p=1}(n) = \mathbb{E}\,\mathrm{tr}(c I_d + W_n)^{-1}.$$

The description replaces baseline precision a_0^{-1} with c. Its predictive value is not, however, determined by this precision difference alone.

Proposition 4.1 (Reliable descriptions)**.** *Let* $p = 1$.

(i) With $r \in (0, 1]$ *fixed,* $\mathrm{ESS}_0 \geq (1 - r)d$*, and* $\mathrm{ESS}_0 \to (1 - r)d$ *as* $a_0 \to \infty$.

(ii) With positive integer c *fixed,* $\mathrm{ESS}_0 \to c + d + 1$ *as* $a_0 \to \infty$.

(iii) With d*,* a_0*, and* c *fixed,* $\mathrm{ESS}_n \to c - a_0^{-1}$ *as* $n \to \infty$.

Part (ii) follows from the two facts of Section 2.4. Without demonstrations, the description's risk is d/c. Under a diffuse baseline prior, $n \geq d+2$ demonstrations leave risk $\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = d/(n-d-1)$, which equals d/c at $n = c+d+1$; integer c places this crossing on the grid used to interpolate the learning curve. Matching information instead of error gives a smaller count. The expected information $\mathbb{E}W_n = nI_d$ of n demonstrations suggests risk d/n, and d/n equals d/c at $n = c$. In this limit, c is also the count that an information-matching ESS, such as that of Morita et al. (2008), assigns to the description (Section 2.5). We call the gap of $d+1$ demonstrations between the two counts the *dimension term*. It reflects the cost of learning a d-dimensional task from random inputs, and it is substantial for a broad description, whereas precision dominates for a specific one. This interpretation concerns the stated limits, rather than an exact additive formula for every finite a_0 and n.

Part (i) fixes the fraction $1-r$ of baseline variance that the description removes, so the description's precision $c = (ra_0)^{-1}$ decreases as a_0 grows. With fewer than d demonstrations, the inputs leave some directions unconstrained, and as $a_0 \to \infty$ the error in those directions dominates. In the limit, the description then replaces $(1-r)d$ demonstrations; at finite a_0, it replaces at least that many.

At $d = 16$ and $a_0 = 10$, a reliable description with $c = 1$ has ESS 15.6, compared with the lower bound $(1-r)d = 14.4$. Increasing c to 100 raises ESS to approximately 117 (Figure 5.1). With many demonstrations, the dimension term vanishes: by part (iii), ESS approaches $c - a_0^{-1}$, the precision that the description adds to the baseline prior. After 100 demonstrations, the $c = 1$ description has ESS 1.1, close to its added precision of 0.9.

The rest of this section proves Proposition 4.1, starting from the properties of W_n that the proof needs.

Notation. Write $\varepsilon = a_0^{-1}$ for the baseline prior precision. For a prior precision $\kappa > 0$, define the Bayes-optimal risk

$$R^{\kappa}(n) = \mathbb{E}\,\mathrm{tr}(\kappa I_d + W_n)^{-1}.$$

Thus $R^{\mathrm{demo}} = R^{\varepsilon}$ and $R^{\mathrm{desc}}_{p=1} = R^{c}$.

Wishart identities. We use three standard properties of W_n. The first two give the facts of Section 2.4.

(W1) If $n < d$, then W_n has rank n almost surely. Consequently, $(\kappa I_d + W_n)^{-1}$ has $d - n$ eigenvalues equal to $1/\kappa$.

(W2) If $n \geq d$ and e is a fixed unit vector, then $(e^\top W_n^{-1} e)^{-1} \sim \chi^2_{n-d+1}$ (Muirhead, 1982, Theorem 3.2.12). Hence

$$\mathbb{E}\,\mathrm{tr}\, W_n^{-1} = \frac{d}{n-d-1} \quad (n \geq d+2),$$

whereas this expectation is infinite for $n = d$ and $n = d+1$.

(W3) For fixed d, as $n \to \infty$,

$$\mathbb{E}\,\mathrm{tr}\, W_n^{-2} = \frac{d}{n^2} + O(n^{-3}), \qquad \mathbb{E}\,\mathrm{tr}\, W_n^{-3} = O(n^{-3})$$

(von Rosen, 1988).

The Sherman–Morrison identity implies that an additional demonstration strictly decreases the expected trace of the posterior covariance. Thus $R^\kappa(n)$ is strictly decreasing, and (W2) implies that it tends to zero. Its linear interpolation is therefore continuous and strictly decreasing.

Large-sample expansion. For fixed d and κ, commuting W_n with κI_d gives

$$(\kappa I_d + W_n)^{-1} = W_n^{-1} - \kappa W_n^{-2} + \kappa^2 W_n^{-2} (\kappa I_d + W_n)^{-1}.$$

The final term is positive semidefinite and bounded above by $\kappa^2 W_n^{-3}$. Using (W2) and (W3),

$$R^\kappa(n) = \frac{d}{n-d-1} - \frac{\kappa d}{n^2} + O(n^{-3}). \tag{4.1}$$

An additional demonstration reduces the leading term by $d/n^2 + O(n^{-3})$. A fixed increase in prior precision and a fixed increase in the demonstration count therefore reduce risk at the same asymptotic order. This comparison gives the large-sample limits in Propositions 4.1(iii) and 4.2(ii), and in Corollary 5.2(ii) for unreliable descriptions.

Proof of Proposition 4.1. A reliable description without demonstrations has risk

$$R^{\mathrm{desc}}_{p=1}(0) = \frac{d}{c} = r a_0 d.$$

(i) Fixed variance reduction. By (W1), $R^{\varepsilon}(n) \geq (d-n)a_0$ for $n < d$; the inequality also holds trivially for $n \geq d$. Linear interpolation preserves this bound, so

$$R^{\mathrm{demo}}(t) \geq (d-t)a_0 \qquad (t \geq 0).$$

For $t < (1-r)d$, the right-hand side exceeds $ra_0 d$. The matching sample size must therefore satisfy $\mathrm{ESS}_0 \geq (1-r)d$.

For every integer n,

$$\frac{R^{\varepsilon}(n)}{a_0} = \mathbb{E}\,\mathrm{tr}(I_d + a_0 W_n)^{-1} \longrightarrow (d-n)^+ \qquad \text{as } a_0 \to \infty.$$

Indeed, an eigenvalue $1/(1+a_0\lambda)$ is bounded by one and converges to the indicator of $\lambda = 0$, so dominated convergence and the almost-sure rank of W_n apply. The same limit holds at every real t by interpolation. For fixed $r \in (0,1)$, the limiting curve crosses rd at the unique point $t = (1-r)d$. Strict monotonicity then gives convergence of the crossing. For $r = 1$, the description is uninformative and $\mathrm{ESS}_0 = 0$ for every a_0, giving the same conclusion.

(ii) Fixed precision. For fixed $c > 0$, the description risk d/c is independent of a_0. As $a_0 \to \infty$, monotone convergence, (W1), and (W2) give

$$R^{\varepsilon}(n) \longrightarrow \begin{cases} +\infty, & n \leq d+1, \\ \dfrac{d}{n-d-1}, & n \geq d+2. \end{cases}$$

If c is a positive integer, the finite part of this limiting curve equals d/c at the integer $n = c+d+1$. For every fixed point below this crossing the limiting risk is larger than d/c, and for every fixed point above it the limiting risk is smaller. The crossing of the interpolated finite-a_0 risk therefore converges to $c+d+1$. The integer condition ensures that the equality occurs at a grid point; linear interpolation does not in general preserve the same expression for noninteger c.

(iii) Many demonstrations. Fix d, a_0, and c. Equation (4.1) implies, uniformly over integers $0 \leq t \leq T = \lceil c \rceil + 1$,

$$R^{\varepsilon}(n+t) - R^{c}(n) = \frac{d}{n^2}\left[c - \varepsilon - t + O(n^{-1})\right].$$

Linear interpolation gives the same expansion for real $t \in [0,T]$. The baseline risk therefore reaches $R^c(n)$ at sample size $n + c - \varepsilon + O(n^{-1})$, so

$$\mathrm{ESS}_n \longrightarrow c - \varepsilon = (1-r)c.$$

□

4.2 Partial Descriptions

Descriptions often specify only part of a task. In our model, a reliable description may constrain only $k < d$ directions. It leaves a fraction r of the baseline variance in these k directions, and it leaves the other $d - k$ directions unchanged. Its prior precision is then c in the constrained directions and ε elsewhere. Write

$$\Lambda = \operatorname{diag}(cI_k, \varepsilon I_{d-k}).$$

The prediction risk is $\mathbb{E}\,\operatorname{tr}(\Lambda + W_n)^{-1}$. Rotational invariance of W_n means that the result depends on the number of constrained directions, not on their orientation.

Proposition 4.2 (ESS of a partial description)**.** *Let $p = 1$ and let $0 \le k < d$ be an integer.*

(i) As $a_0 \to \infty$, $\mathrm{ESS}_0 \to (1 - r)k$ with r fixed, and $\mathrm{ESS}_0 \to k$ with c fixed.

(ii) For fixed d, a_0, and c, as $n \to \infty$,

$$\mathrm{ESS}_n \longrightarrow \frac{k}{d}(c - a_0^{-1}) = \frac{k}{d}(1 - r)c.$$

Without demonstrations, the unconstrained directions dominate the remaining error, so with c fixed the ESS approaches k. With many demonstrations, it approaches the limit in part (ii), the added precision averaged across all d directions. This limit exceeds k when, approximately, $c - a_0^{-1} > d$. A partial description this precise gains value as demonstrations resolve what it leaves unspecified. At $d = 16$, $k = 8$, and $c = 100$, ESS rises from 8.1 without demonstrations to 36.2 after 100 demonstrations (Table 4.1). Over the same range, the ESS of a complete description at $c = 100$ falls from 117 to 109, so demonstrations substitute for a complete description and complement a partial one. At $c = 1$, ESS falls as demonstrations accumulate, whether the description is partial or complete.

Proof of Proposition 4.2. (i) No demonstrations. The description risk is

$$\operatorname{tr}\Lambda^{-1} = (kr + d - k)a_0.$$

Using $R^{\mathrm{demo}}(t) \ge (d - t)a_0$ gives $\mathrm{ESS}_0 \ge (1 - r)k$. The normalized baseline risk converges to $(d-t)^+$, locally uniformly in t, by the integer limits and linear interpolation. With r fixed, the normalized description risk is $d - (1 - r)k$. With c fixed, it is

$$d - k + \frac{k}{ca_0} \longrightarrow d - k.$$

Table 4.1: **Demonstrations can increase the value of a precise partial description.** Bayes-optimal ESS for a reliable description of k of $d = 16$ directions, with $a_0 = 10$. The case $k = 16$ is a complete description. Each block reports ESS after n demonstrations and its large-sample limit $(1 - r)ck/d$. Estimates use eight batches of 2,000 prompts; standard errors are at most 0.05.

	$c = 1$					$c = 100$				
k	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$
1	0.9	0.8	0.1	0.1	0.1	1.0	1.0	3.5	5.2	6.2
4	3.6	2.9	0.3	0.2	0.2	4.0	4.2	15.5	21.7	25.0
8	7.3	5.1	0.5	0.5	0.5	8.1	9.8	36.2	46.0	50.0
12	11.1	6.7	0.8	0.7	0.7	12.4	23.4	65.1	73.0	74.9
16	15.6	8.0	1.1	0.9	0.9	117	116	109	103	99.9

Because $k < d$, both limiting levels are positive and have unique crossings, at $(1 - r)k$ and k, respectively. Convergence of the crossings proves the claim. When the description leaves the prior unchanged, its ESS is identically zero.

(ii) Many demonstrations. Applying the resolvent identity twice gives

$$(\Lambda + W_n)^{-1} = W_n^{-1} - W_n^{-1}\Lambda W_n^{-1} + W_n^{-1}\Lambda W_n^{-1}\Lambda(\Lambda + W_n)^{-1}.$$

Rotational invariance implies

$$\mathbb{E}W_n^{-2} = \frac{\mathbb{E}\,\mathrm{tr}\,W_n^{-2}}{d} I_d.$$

The expected trace of the middle term is therefore $\mathrm{tr}(\Lambda)/n^2 + O(n^{-3})$ by (W3). To bound the remainder, use $|\,\mathrm{tr}(MB)| \le \|M\|_{\mathrm{op}}\,\mathrm{tr}\,B$ for $B \succeq 0$, together with $(\Lambda + W_n)^{-1} \preceq W_n^{-1}$. The remainder's absolute trace is at most

$$\|\Lambda\|_{\mathrm{op}}^2 \|W_n^{-1}\|_{\mathrm{op}}^2\,\mathrm{tr}\,W_n^{-1} \le d\|\Lambda\|_{\mathrm{op}}^2\,\mathrm{tr}\,W_n^{-3},$$

whose expectation is $O(n^{-3})$. Consequently,

$$\mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1} = \frac{d}{n - d - 1} - \frac{\mathrm{tr}\,\Lambda}{n^2} + O(n^{-3}).$$

Matching this expansion to the baseline risk, as in the proof of Proposition 4.1(iii), gives

$$\mathrm{ESS}_n \longrightarrow \frac{\mathrm{tr}\,\Lambda - d\varepsilon}{d} = \frac{k(c - \varepsilon)}{d}.$$

□

The same expansion applies to a Gaussian description with any fixed precision matrix $\Lambda \succeq \varepsilon I_d$ compatible with the baseline task distribution. The large-sample ESS of such a description is the added precision averaged across directions, $(\operatorname{tr} \Lambda - d\varepsilon)/d$.

Chapter 5

The Cost of Unreliability

We now allow the description to be irrelevant with probability $1 - p$. Uncertain relevance limits the benefit of increasing precision: for every fixed $p < 1$, the zero-demonstration risk remains positive as $c \to \infty$, giving a finite ESS ceiling. Section 5.1 derives the risk and ESS of the Bayes-optimal predictor for an unreliable description; with demonstrations, this predictor also infers whether the description is relevant. Section 5.2 shows that a predictor that assumes relevance can have higher risk than the demonstrations-only predictor.

5.1 Unreliable Descriptions

For $0 < p < 1$, let $\mu_z = \mathbb{E}[w \mid m, D_n, Z = z]$ be the component posterior means and let $\pi_n = P(Z = 1 \mid m, D_n)$ be the *posterior relevance probability*. Bayesian model averaging (Hoeting et al., 1999) gives

$$\hat{y} = \big[\pi_n \mu_1 + (1 - \pi_n)\mu_0\big]^\top x.$$

Demonstrations affect both the component predictions and their relative weight. The law of total variance (2.6) then yields the following risk decomposition.

Proposition 5.1 (Risk under uncertain relevance)**.** *For every $n \geq 0$,*

$$\begin{aligned} R^{\text{desc}}(n) = {} & p R^{\text{desc}}_{p=1}(n) \\ & + (1 - p) R^{\text{demo}}(n) + \mathcal{V}_n, \end{aligned} \tag{5.1}$$

where

$$\mathcal{V}_n = \mathbb{E}\Big[\pi_n(1 - \pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2\Big] \geq 0.$$

Without demonstrations,

$$R^{\text{desc}}(0) = \big[1 - p^2(1-r)\big] a_0 d. \tag{5.2}$$

Here $\mathcal{V}_n$ is the expected between-component posterior variance. The first two terms in Equation (5.1) are the risk of an *oracle*, the Bayes-optimal predictor that also observes Z, so $\mathcal{V}_n$ is the excess Bayes risk due to unobserved relevance. The oracle uses the description when it is relevant and otherwise uses the baseline prior; $\mathcal{V}_n$ is the disagreement between these two predictions, weighted by uncertainty about which is appropriate. At $n = 0$, a predictor that observed Z would remove a fraction $p(1-r)$ of baseline risk. The Bayes-optimal predictor does not observe Z: it shrinks m by the factor p and removes only $p^2(1-r)$.

The cost of uncertainty about relevance is small when the two predictions agree. It is also small when the demonstrations make one component far more likely than the other, because $\pi_n(1-\pi_n)$ is then near zero. Without demonstrations, the cost is $\mathcal{V}_0 = p(1-p)(1-r)a_0 d$, which grows as a more specific description pulls the two predictions apart. With demonstrations, however, a specific description is also easier to distinguish from an irrelevant one.

Corollary 5.2 (ESS under uncertain relevance)**.**

(i) For $0 < p < 1$, define $n_p^\sharp$ by

$$R^{\text{demo}}(n_p^\sharp) = (1-p^2) R^{\text{demo}}(0).$$

Then $\text{ESS}_0 < n_p^\sharp$ at every finite c, and $\text{ESS}_0 \to n_p^\sharp$ as $c \to \infty$. As $a_0 \to \infty$ with r fixed, $\text{ESS}_0 \to p^2(1-r)d$ and $n_p^\sharp \to p^2 d$. For $p = 0$, both ESS and the ceiling are zero.

(ii) For $p < 1$, with d, a_0, and c fixed,

$$\limsup_{n\to\infty} \text{ESS}_n \le p(c - a_0^{-1}),$$

p times the large-sample ESS of a reliable description (Proposition 4.1(iii)).

Without demonstrations, uncertain relevance imposes an error floor. For fixed $p < 1$, the risk in Equation (5.2) approaches the positive floor $(1-p^2)a_0 d$ as $c \to \infty$, giving a finite ceiling on zero-demonstration ESS. We take $p = 0.9$, the weight

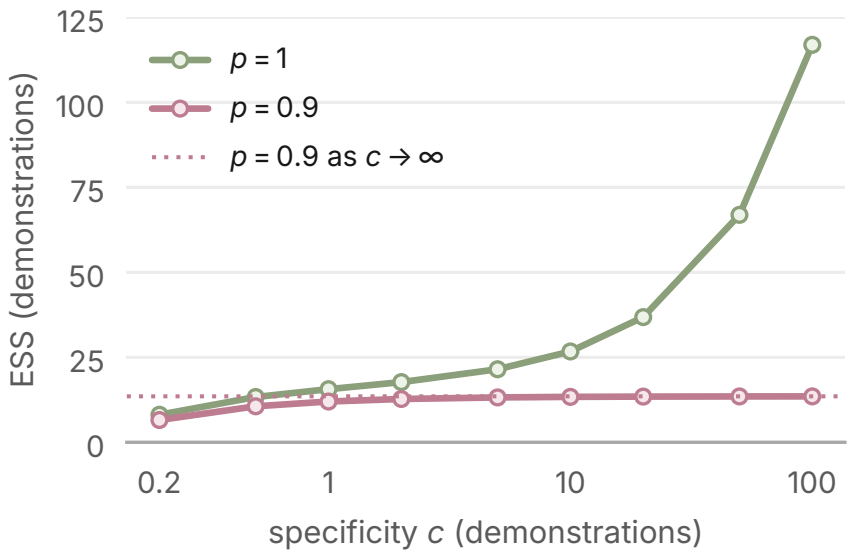


Figure 5.1: **Without demonstrations, reliability limits the benefit of specificity.** Bayes-optimal ESS_0 versus precision c, for $p = 1$ and $p = 0.9$, with $d = 16$ and $a_0 = 10$. The dotted line is the finite ceiling as $c \to \infty$ at $p = 0.9$. Estimates use eight batches of 4,000 prompts; standard errors are below 0.02 demonstrations.

on the informative component in the robust-prior example of Schmidli et al. (2014). With $d = 16$ and $a_0 = 10$, increasing c from 1 to 100 then raises ESS only from 12.0 to 13.5, approximately the ceiling. At $p = 1$, the same increase raises ESS from 15.6 to 117 (Figure 5.1). As the baseline prior becomes diffuse, the ceiling tends to $p^2 d = 12.96$.

The oracle has the lower floor $(1-p)a_0 d$, the risk of a reliable description that leaves a fraction $1 - p$ of the baseline variance. In this setting, the oracle's ESS ceiling is therefore 15.6 demonstrations, the ESS of the reliable $c = 1$ description. Not observing relevance lowers the ceiling to about 13.5.

With many demonstrations, ESS is at most p times that of a reliable description. Demonstrations reduce $R^{\text{demo}}(n)$, the risk when the description is irrelevant, so the term $(1-p)R^{\text{demo}}(n)$ in Equation (5.1) falls and ESS can grow. In the limit, a reliable description is worth $c - a_0^{-1}$ demonstrations (Proposition 4.1(iii)), and an unreliable one at most p times as much (Corollary 5.2(ii)). After 100 demonstrations, the $c = 100$ description has ESS 109 when $p = 1$ and 86.6 when $p = 0.9$, compared

with limits of 99.9 and at most 89.9. The reliable value exceeds its limit because the dimension term has not yet vanished at $n = 100$.

Demonstrations distinguish low precision from uncertain relevance. Without demonstrations, an unreliable description has the same risk as a reliable description that leaves a fraction $1 - p^2(1 - r)$ of the baseline variance. For example, a $c = 100$ description with $p = 0.9$ and a reliable one with $c \approx 0.52$ both have ESS about 13.5 here. After 100 demonstrations, their ESS values are 86.6 and 0.5. Demonstrations can provide evidence for the high-precision component of the unreliable description; the reliable low-precision description has no such component.

The rest of this section proves Proposition 5.1 and Corollary 5.2.

Component posteriors. Let $\mathcal{P}_n = (D_n, m, x)$ denote the full prompt, including the query input. Conditional on $Z = 1$, the posterior parameters are

$$\Sigma_1 = (cI_d + W_n)^{-1}, \qquad \mu_1 = \Sigma_1(cm + X_n^\top y_{1:n}).$$

Conditional on $Z = 0$, the description center m is independent of w, and

$$\Sigma_0 = (\varepsilon I_d + W_n)^{-1}, \qquad \mu_0 = \Sigma_0 X_n^\top y_{1:n}.$$

The posterior relevance probability follows from the two Gaussian marginal likelihoods,

$$\begin{aligned} L_1 &= \mathcal{N}\big(y_{1:n}; X_n m, I_n + c^{-1} X_n X_n^\top\big)\,, \\ L_0 &= \mathcal{N}\big(y_{1:n}; 0, I_n + a_0 X_n X_n^\top\big)\,, \\ \pi_n &= \frac{pL_1}{pL_1 + (1-p)L_0}. \end{aligned}$$

The query input does not enter this update because it is independent of (Z, m, D_n), so $\pi_n = P(Z = 1 \mid \mathcal{P}_n)$. Without demonstrations, both marginal likelihoods equal one and $\pi_0 = p$, since m has the same distribution under both relevance states. At $p = 0$ or $p = 1$, the posterior contains only the corresponding component. At $r = 1$, $m = 0$ almost surely and the two components coincide, so the description provides no additional task information for any p.

Proof of Proposition 5.1. The posterior mean prediction averages the component predictions using their posterior probabilities:

$$\mathbb{E}[w^\top x \mid \mathcal{P}_n] = \big(\pi_n \mu_1 + (1 - \pi_n)\mu_0\big)^\top x.$$

The law of total variance (2.6) over Z separates posterior variance into within-component and between-component terms:

$$\begin{aligned}\mathrm{Var}(w^\top x \mid \mathcal{P}_n) = \pi_n x^\top \Sigma_1 x + (1-\pi_n) x^\top \Sigma_0 x \\ + \pi_n(1-\pi_n)\big((\mu_1-\mu_0)^\top x\big)^2.\end{aligned}$$

For $0 < p < 1$ and any integrable function g of the prompt,

$$\mathbb{E}[\pi_n g(\mathcal{P}_n)] = \mathbb{E}[Z g(\mathcal{P}_n)] = p\,\mathbb{E}[g(\mathcal{P}_n) \mid Z = 1].$$

Conditional on $Z = 1$, the prompt follows the reliable-description model. Conditional on $Z = 0$, the description provides no information about the task. Taking expectations in the variance decomposition therefore gives

$$R^{\mathrm{desc}}(n) = pR^{\mathrm{desc}}_{p=1}(n) + (1-p)R^{\mathrm{demo}}(n) + \mathcal{V}_n,$$

with $\mathcal{V}_n$ as in Equation (5.1). The endpoint cases $p = 0$ and $p = 1$ follow directly from the single-component posteriors.

At $n = 0$, $\mu_1 = m$, $\mu_0 = 0$, and the prediction is $p\,m^\top x$. For each coordinate,

$$\mathrm{Var}(w_i) = a_0, \qquad \mathrm{Var}(m_i) = (1-r)a_0, \qquad \mathrm{Cov}(w_i, m_i) = p(1-r)a_0.$$

The covariance follows because, conditional on relevance, w_i is m_i plus independent noise, and conditional on irrelevance, w_i is independent of m_i. It follows that

$$\mathbb{E}(w_i - pm_i)^2 = a_0 - 2p^2(1-r)a_0 + p^2(1-r)a_0 = \big(1 - p^2(1-r)\big)a_0.$$

Since $\mathbb{E}[xx^\top] = I_d$, the prediction risk is $\mathbb{E}\|w - pm\|^2$. Summing over coordinates proves Equation (5.2). □

Proof of Corollary 5.2. (i) No demonstrations. For $0 < p < 1$ and finite c,

$$R^{\mathrm{desc}}(0) = \big(1 - p^2(1-r)\big)R^{\mathrm{demo}}(0) > (1-p^2)R^{\mathrm{demo}}(0).$$

The risk on the left decreases to the right-hand side as $c \to \infty$, equivalently $r \to 0$. The interpolated baseline risk is continuous, strictly decreasing, and tends to zero. Consequently, the matching sample size is strictly below the unique finite $n_p^\sharp$ defined in the corollary and converges to it as $c \to \infty$. At $p = 0$, the description is always irrelevant, so $\mathrm{ESS}_0 = n_0^\sharp = 0$; the non-strict bound remains valid.

The baseline lower bound and normalized limit established in the proof of Proposition 4.1(i) also give

$$\mathrm{ESS}_0 \geq p^2(1-r)d, \qquad \mathrm{ESS}_0 \longrightarrow p^2(1-r)d$$

when $a_0 \to \infty$ with r fixed. Applying the same argument to the level $1-p^2$ gives $n_p^\sharp \to p^2 d$. These limits need not hold exactly at finite a_0.

For comparison, the oracle that observes Z has zero-demonstration risk

$$\big(1-p(1-r)\big)a_0 d \geq (1-p)a_0 d.$$

The oracle's zero-demonstration ESS ceiling therefore approaches pd as $a_0 \to \infty$. For the learner that must infer relevance, this limit is $p^2 d$. The difference is the cost of uncertainty about relevance.

(ii) Many demonstrations. With many demonstrations, an additional demonstration reduces risk by about d/n^2, and a prior of precision κ reduces it by κ times as much (Equation (4.1)). A reliable description, which raises the precision from ε to c, therefore saves about $c-\varepsilon$ demonstrations (Proposition 4.1(iii)). A description that is relevant with probability p saves about p times as much, and the term $\mathcal{V}_n$ can only lower its value further.

Fix d, a_0, and c, and define the oracle risk

$$L(n) = pR^c(n) + (1-p)R^\varepsilon(n).$$

As in the proof of Proposition 4.1(iii), Equation (4.1) implies, uniformly over integers $0 \leq t \leq T = \lceil c \rceil + 1$,

$$R^{\mathrm{demo}}(n+t) - L(n) = \frac{d}{n^2}\left[p(c-\varepsilon) - t + O(n^{-1})\right].$$

Linear interpolation gives the same expansion for real $t \in [0,T]$. Let $n+\eta_n$ be the sample size at which the baseline risk equals $L(n)$. Then

$$\eta_n = p(c-\varepsilon) + O(n^{-1}).$$

When $p(c-\varepsilon) = 0$, the two risks coincide and $\eta_n = 0$ exactly. For $p < 1$, the variance decomposition gives $R^{\mathrm{desc}}(n) = L(n) + \mathcal{V}_n \geq L(n)$. Since the baseline risk decreases with sample size, $\mathrm{ESS}_n \leq \eta_n$, proving

$$\limsup_{n\to\infty} \mathrm{ESS}_n \leq p(c-\varepsilon) = p(1-r)c.$$

Because $\mathcal{V}_n$ is nonnegative and need not vanish at the required rate, the result for unreliable descriptions is an upper bound. □

5.2 Assuming Relevance Under Misspecification

A learner that has only ever seen relevant descriptions has no reason to discount a description; a network trained at $p = 1$ is such a learner (Chapter 6). This section asks what assuming relevance costs when descriptions are in fact unreliable. Consider a Bayesian predictor that assumes relevance by setting $p = 1$ in its inference model. It therefore always returns $\mu_1^\top x$, even when the description is irrelevant. We evaluate its risk, denoted $R^{\text{assume}}(n)$, under the data-generating model with relevance probability p. The assumed prior is misspecified when $p < 1$ and $r < 1$, because it omits the irrelevant component.

Without demonstrations, this predictor returns $m^\top x$. By the moments in the proof of Proposition 5.1, each coordinate has $\mathbb{E}(w_i - m_i)^2 = a_0 - 2p(1-r)a_0 + (1-r)a_0$, so

$$R^{\text{assume}}(0) = \mathbb{E}\|w - m\|^2 = \big(1 - (2p-1)(1-r)\big)a_0 d.$$

For $r < 1$, this risk exceeds the demonstrations-only risk $a_0 d$ exactly when $p < 1/2$. At $r = 1$, the description carries no information, $m = 0$ almost surely, and the two risks coincide for every p.

With demonstrations, the predictor continues to use $\mu_1^\top x$ regardless of Z, where

$$\mu_1 = m + (cI_d + W_n)^{-1} X_n^\top (y_{1:n} - X_n m).$$

For fixed a_0 and n, $\mu_1 \to m$ in mean square as $c \to \infty$. Conditional on relevance, $\mathbb{E}\|w - m\|^2 = r a_0 d \to 0$; conditional on irrelevance, it is $(2 - r)a_0 d \to 2a_0 d$. Thus

$$\lim_{c \to \infty} R^{\text{assume}}(n) = 2(1-p)a_0 d.$$

For every $p < 1$, this is a positive risk floor at fixed n, whereas $R^{\text{demo}}(n) \to 0$ as $n \to \infty$. A sufficiently precise, occasionally irrelevant description can therefore outweigh useful demonstrations. Take $p = 0.9$, $c = 100$, $d = 16$, and $a_0 = 10$. At $n = 14$, the predictor assuming relevance has risk 25.9, compared with 26.6 without the description. At $n = 15$, these risks are 25.5 and 19.5. The risk of the predictor assuming relevance remains higher at every larger demonstration count evaluated, up to 10^5. The order of limits matters: the preceding argument takes $c \to \infty$ at fixed n; it does not imply that demonstrations cannot overcome a fixed finite prior precision.

Chapter 6

Meta-Trained Transformers

Transformers meta-trained on demonstrations can approach Bayes-optimal prediction (Garg et al., 2022; Genewein et al., 2025). We first test whether they also attain the Bayes-optimal ESS of descriptions under the correctly specified model. Following Garg et al. (2022), we train them from scratch, with $a_0 = 10$ as before. The dimension is $d = 5$ instead of 16: this is the dimension used by Huang & Ge (2025), and it keeps training short. Prompts contain an optional description token carrying m, up to 20 demonstration tokens (40 at $d = 10$), and a query token (Huang & Ge, 2025). The description token does not state c or p; the network must learn these properties across training tasks. Half the prompts omit the description; a network's predictions on such prompts give its own baseline curve. We train separate networks for $c \in \{1, 10\}$ and $p \in \{1, 0.99, 0.9\}$, with two or three seeds per configuration. The two precisions are a factor of ten apart, and so are the two nonzero probabilities of irrelevance, $1 - p = 0.01$ and $1 - p = 0.1$. We omit $c = 100$: such a description is worth more demonstrations than a 20-demonstration prompt can show. We train further networks with $c \in \{0.2, 2\}$, with $d = 10$, and with a partial description. Section 6.1 gives the training and evaluation details. Section 6.2 compares the networks with the Bayes-optimal predictor when training and test reliability match, and Section 6.3 tests them when the two differ.

6.1 Training and Evaluation

Architecture and optimization. The Transformer has 12 layers, 8 attention heads, and width 256, matching the model size of Garg et al. (2022). Layers use pre-normalization, GELU activations, feed-forward width 1,024, and no dropout. At $d = 5$, the model has 9.5 million parameters. Unlike the model of Garg et al. (2022), ours uses bidirectional attention and no positional encoding; two indicator coordinates distinguish token types. Each demonstration token contains an input and its label. When a description is absent, its token is masked out of attention. The prediction is read from the query token.

Following Zhu et al. (2026), the model predicts a mean and a log variance for the query label and minimizes log loss. The optimal predicted mean is then the posterior mean, which also minimizes squared error (Section 2.4), so we evaluate the predicted mean by the prediction error of Equation (3.1).

We train with Adam, the optimizer of Garg et al. (2022), at learning rate 10^{-4} for 40,000 steps, using batches of 1,024 newly sampled prompts and no curriculum. Adam uses the PyTorch defaults $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$, with no weight decay and a constant learning rate. Labels and description centers are divided by $\sqrt{a_0}$, gradients are clipped to norm 1, and training uses bf16 autocast. The demonstration count is sampled uniformly from 0 to 20, except for the $c = 0.2$ and $c = 2$ experiments, which use 0 to 15, and the $d = 10$ experiments, which use 0 to 40. Training seeds are 0, 1, and 2 for $c = 10$, for $d = 10$, and for the partial-description experiments; 0 and 1 for $c = 1$; and 0 for $c \in \{0.2, 2\}$. A $d = 5$ network takes approximately 80 minutes to train on one NVIDIA L4 GPU, and a $d = 10$ network takes approximately 2.8 hours. Training all 25 networks takes approximately 38 GPU-hours.

Evaluation. We evaluate each network on 20,000 fresh prompts. The networks share these evaluation prompts, so standard deviations across training seeds exclude prompt-sampling variation. Each network with a complete description and $c \in \{1, 10\}$ is tested at its training reliability and at $p = 1$ and $p = 0.9$; those with seeds 1 and 2 are also tested at $p = 0.99$. At $d = 5$, we evaluate the baseline curve at every integer from 0 to 20, except for $c = 0.2$ and $c = 2$, which use counts from 0 to 15. We compute ESS at the first crossing of this curve with the prediction error obtained using the description, as in Definition 3.1. Figures 6.1a and 6.2 compare each network with the

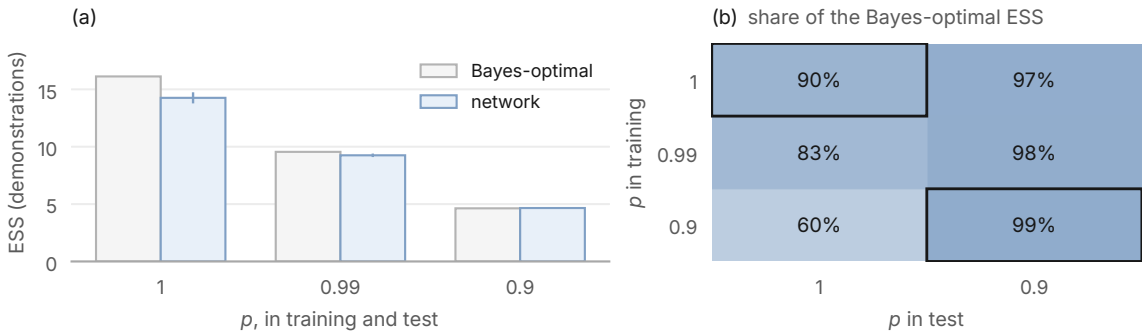


Figure 6.1: **Meta-trained Transformers approach Bayes-optimal ESS and reflect training reliability.** Results use $c = 10$, $d = 5$, $a_0 = 10$, and no demonstrations. (a) Training and test reliability match. Network ESS uses each network's own baseline curve; bars show the mean and one standard deviation across three training seeds. The Bayes-optimal comparison uses the same prompts. (b) Training and test reliability vary separately. Cells show network ESS as a fraction of the Bayes-optimal ESS under the test distribution; outlined cells have matching reliabilities. Table 6.1 includes $c = 1$.

Bayes-optimal predictor on the same prompts. Table 6.1 reports results at $n = 0$, including test reliabilities that differ from the training reliability, with Bayes-optimal values estimated on 200,000 prompts.

6.2 Matched Training and Test Reliability

Because ESS is measured against each network's own baseline curve, we first check that the networks learn the task from demonstrations alone. Without a description, every $d = 5$ network's error is within 0.21 of the Bayes-optimal error of approximately 50 at $n = 0$, and within 0.32 on average over its training demonstration counts. At $d = 10$, where the Bayes-optimal error at $n = 0$ is approximately 100, the gap is at most 0.50 at $n = 0$ and at most 0.41 on average. With a description at $c = 10$, $p = 1$, and $n = 0$, network error ranges from 0.67 to 0.71, compared with 0.49 for the Bayes-optimal predictor. The networks therefore remove 98.6% of the baseline error, compared with 99.0% for the Bayes-optimal predictor.

ESS expresses this remaining gap in units of demonstrations. With no demonstrations and $c = 10$, network ESS is 14.3 ± 0.5 at $p = 1$, 9.3 ± 0.2 at $p = 0.99$, and

4.66 ± 0.01 at $p = 0.9$, compared with Bayes-optimal values of 16.1, 9.6, and 4.63 on the same prompts (Figure 6.1a). A network's ESS can exceed the Bayes-optimal ESS on the same prompts, as it does at $p = 0.9$, because the network's ESS is measured against its own baseline curve, not the Bayes-optimal curve. On 200,000 prompts, the Bayes-optimal value at $p = 1$ is 15.9, close to the limiting expression $c + d + 1 = 16$ of Proposition 4.1. At $p = 0.99$, the Bayes-optimal value is 8.96 on 200,000 prompts, below the networks' mean ESS of 9.3, and 9.6 on the networks' evaluation prompts, above that mean. For $c = 1$, network ESS is within 0.1 of the Bayes-optimal ESS at every reliability (Table 6.1); at $p = 1$, for example, it is 5.85 ± 0.03, compared with 5.89.

To check that this agreement does not depend on the two precisions above, we train four further networks, with $c = 0.2$ and $c = 2$, each trained and tested at $p = 1$ and at $p = 0.9$. At $n = 0$, each comes within 0.03 of the Bayes-optimal ESS on the same prompts. At $c = 0.2$, the network and Bayes-optimal values are 2.64 and 2.66 for $p = 1$, and 2.16 and 2.15 for $p = 0.9$. At $c = 2$, they are 7.32 and 7.35 for $p = 1$, and 4.38 and 4.37 for $p = 0.9$.

To test whether the networks also show the dimension term of Proposition 4.1, we train three seeds at $d = 10$, with $c = 10$ and $p = 1$, and extend training and evaluation to 40 demonstrations so that the baseline curve reaches the ESS crossing. Network ESS is 17.8, with standard deviation 0.3 across seeds, compared with a Bayes-optimal value of 20.8 and the limiting expression $c + d + 1 = 21$. As at $d = 5$, the networks capture most of the description's predictive value but fall short of the Bayes-optimal ESS: by 3.0 demonstrations here, compared with 1.8 at $d = 5$.

Figure 6.2 follows ESS as demonstrations accumulate. For $c = 1$, network ESS remains within 0.6 of the Bayes-optimal value at every demonstration count whose crossing lies within the evaluation grid. For $c = 10$, the largest shortfalls are 2.4, 2.9, and 1.4 demonstrations at $p = 1$, 0.99, and 0.9, respectively; the largest excess is 0.3. The shortfall is consistent across seeds: at $n = 5$, the three $c = 10$ networks trained at $p = 0.9$ have ESS values of 4.83, 4.79, and 4.77, all below the Bayes-optimal value of 5.20 on the same prompts.

Networks also learn that a partial description and demonstrations complement each other. The partial-description networks receive a reliable partial description (Section 4.2) that constrains $k = 3$ of the $d = 5$ coordinates with precision $c = 10$. The description token contains m in the first three coordinates and zeros elsewhere,

Table 6.1: **Transformer ESS and fitted description weights under matched and shifted reliability.** Results use $d = 5$, $a_0 = 10$, and no demonstrations, with 20,000 test prompts per network. Rows are grouped by training reliability. *Network* reports mean ESS and standard deviation across the indicated training seeds; for a single seed, the uncertainty is the standard error over prompts. The two Bayesian predictors use p_{train} and p_{test}, respectively; the latter is Bayes-optimal under the test distribution. We show one reference value when they coincide. Seed 0 was not evaluated at $p_{\text{test}} = 0.99$ for training reliabilities 1 and 0.9, so those rows average fewer seeds; their fitted weights differ from the other rows only because of the seeds included. Values using p_{test}, and coincident values using p_{train}, use 200,000 prompts, with standard errors at most 0.1. Other values using p_{train} use the networks' test prompts; sampling variation can change these estimates by up to 0.6, so a Monte Carlo estimate using p_{train} can exceed the corresponding estimate using p_{test}. *Weight* is the fitted description weight q in $\hat{y} \approx q\, m^\top x$ (Section 6.3), averaged across the available seeds. This coefficient is unrestricted and need not be a relevance probability.

				ESS			
c	p_{train}	p_{test}	Seeds	Network	Bayes p_{train}	Bayes p_{test}	Weight q
1	1	1	2	5.85 ± 0.03	5.89	—	0.964
1	1	0.99	1	5.65 ± 0.06	5.66	5.61	0.991
1	1	0.9	2	4.02 ± 0.03	3.98	4.10	0.964
10	1	1	3	14.26 ± 0.48	15.91	—	1.008
10	1	0.99	2	9.38 ± 0.16	9.54	8.96	1.015
10	1	0.9	3	4.58 ± 0.03	4.54	4.70	1.008
1	0.99	1	2	5.87 ± 0.01	5.88	5.89	0.963
1	0.99	0.99	2	5.65 ± 0.01	5.61	—	0.963
1	0.99	0.9	2	4.02 ± 0.03	3.99	4.10	0.963
10	0.99	1	3	13.28 ± 0.57	16.01	15.91	0.982
10	0.99	0.99	3	9.25 ± 0.15	8.96	—	0.982
10	0.99	0.9	3	4.60 ± 0.04	4.56	4.70	0.982
1	0.9	1	2	5.67 ± 0.06	5.75	5.89	0.883
1	0.9	0.99	1	5.51 ± 0.06	5.55	5.61	0.901
1	0.9	0.9	2	4.03 ± 0.01	4.10	—	0.883
10	0.9	1	3	9.49 ± 0.44	10.79	15.91	0.876
10	0.9	0.99	2	8.08 ± 0.28	8.55	8.96	0.880
10	0.9	0.9	3	4.66 ± 0.01	4.70	—	0.876

and the other two task coordinates follow the baseline prior. Their mean ESS rises from 3.27 at $n = 0$ to 5.55 at $n = 10$, compared with Bayes-optimal values 3.22 and 5.64; the standard deviation across the three seeds at $n = 0$ is 0.03. This rise agrees

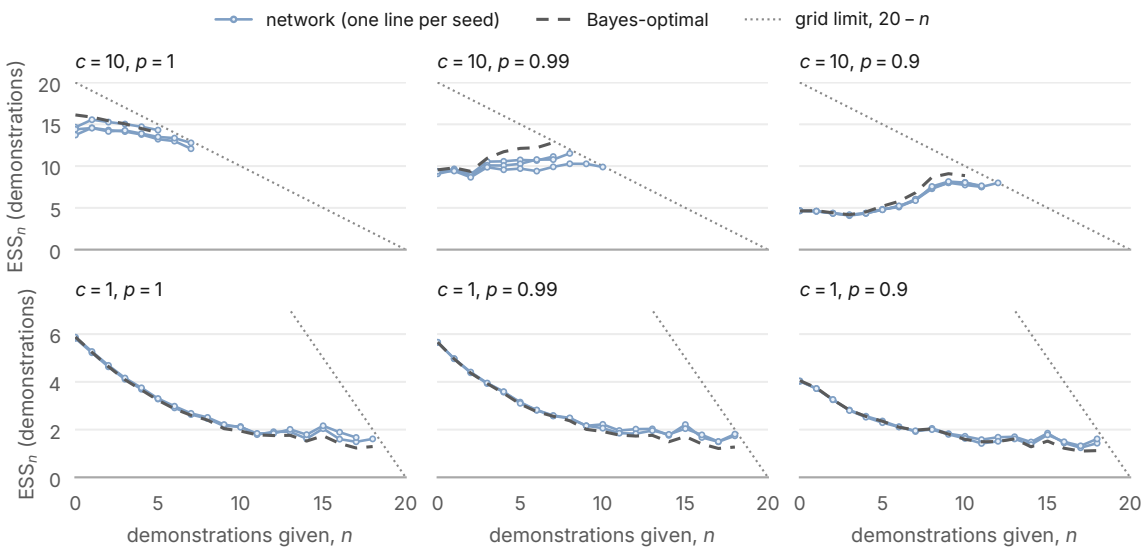


Figure 6.2: **Transformer ESS as a function of demonstration count.** Each line represents a training seed; the Bayes-optimal predictor is evaluated on the same prompts. Training and test reliability are matched. Results cover $c = 10$ and $c = 1$ at $d = 5$, using 20,000 prompts. A curve ends when its crossing exceeds the maximum of 20 demonstrations. The dotted line, $20 - n$, marks the largest ESS identifiable on this grid.

with Proposition 4.2, whose large-sample limit here is $k(c - a_0^{-1})/d = 5.94$.

6.3 Reliability Shifts

The networks learn reliability from their training data, but descriptions at test time may be more or less reliable. We therefore evaluate each network at test reliabilities that differ from its training reliability, and compare it with two Bayesian predictors (Raventós et al., 2023): one that uses the training reliability p_{train}, and one that uses the test reliability p_{test} and is therefore Bayes-optimal. At $n = 0$, a Bayesian predictor with reliability p predicts $p\, m^\top x$. We therefore summarize a network's predictions by the *fitted description weight* q, the coefficient from regressing them on $m^\top x$. This coefficient describes prediction behavior and need not represent a probability inside the network.

Without demonstrations. For $c = 10$, the networks trained at $p = 0.9$ have fitted description weights of 0.87 to 0.89 across the three seeds, and keep them even when every test description is relevant (Figure 6.1b and Table 6.1). Their ESS on reliable descriptions is then 9.5 ± 0.4. The Bayesian predictor that uses p_{train} reaches 10.8 on the same prompts, and the Bayes-optimal predictor under the test distribution reaches 15.9 on 200,000 prompts. This is consistent with the information available: m alone does not reveal the changed test reliability. The networks trained at $p = 1$ or $p = 0.99$ have fitted weights between 0.95 and 1.02, so the two training reliabilities cannot be distinguished without demonstrations. At test reliability $p = 0.9$, assuming relevance costs the networks trained at $p = 1$ little at $n = 0$: their ESS is 4.58 ± 0.03, compared with 4.70 for the Bayes-optimal predictor under the test distribution on 200,000 prompts.

After demonstrations are observed. With demonstrations, the networks trained at $p = 1$ and at $p = 0.99$ respond differently to a test reliability of $p = 0.9$, under which 10% of test descriptions are irrelevant. For the networks trained at $p = 0.99$, the description reduces error at every evaluated n, as it does for the Bayes-optimal predictor under the test distribution. Across seeds, the networks' reduction is 3.5 to 3.7 with six demonstrations and 0.08 to 0.13 with 20. For the networks trained at $p = 1$, the description increases error from six demonstrations onward, in every seed: by 0.77 to 1.06 at $n = 6$ and by 2.74 to 2.94 at $n = 9$, where the excess peaks. With eight demonstrations, for example, the seed-0 network has error 4.6 with the description and 2.3 without it. A Bayesian predictor that assumes relevance (Section 5.2) also first has higher error than the demonstrations-only predictor at $n = 6$, with excess error 1.12 there and 2.96 at $n = 9$. The networks trained at $p = 1$ thus behave as if every description were relevant.

Chapter 7

Pretrained Language Models

Unlike the meta-trained networks, pretrained language models must be told a description's precision and reliability. We ask whether they use what the prompt states. To isolate the use of a description from the ability to carry out linear regression in context, we study numerical mean estimation, which is the scalar case of our model with fixed input $x = 1$. A semantic classification task follows in Section 7.4. The scalar task has an exact Bayesian reference and tests the use of stated precision and relevance probabilities without the dimension term of Chapter 4, which arises from random inputs.

7.1 Task and Evaluation

Each numerical task has a hidden mean $w \sim \mathcal{N}(500, 100^2)$. Its demonstrations are drawn from $\mathcal{N}(w, \sigma^2)$, with $\sigma = 60$, and rounded to integers. Language models predict short unsigned numbers best (Gruver et al., 2023; Requeima et al., 2024), so a task is redrawn if any of its numbers falls outside 100 to 999. A description gives a center m and standard deviation τ: "Its mean was drawn from a normal distribution with mean 510 and standard deviation 30." Before rounding and redrawing, the relevant component satisfies $w \mid m \sim \mathcal{N}(m, \tau^2)$. Its precision, in units of the precision of one demonstration as in Chapter 3, is $c = \sigma^2/\tau^2$; with $x = 1$, each demonstration is a direct observation of the task. Stated standard deviations 60, 30, and 10 correspond to $c = 1$, 4, and 36, so that the reference's weight on the description, $c/(c+n)$, ranges from 0.2 to 0.97 over $n \leq 4$. The noise level $\sigma = 60$ makes this range possible with

three-digit numbers: the smallest stated standard deviation, 10, is still large relative to rounding, and the largest, 60, is small relative to the range 100 to 999.

We vary the stated precision. A more precise description also has a more accurate center, so we add a condition that changes only the statement: each $c = 36$ task keeps its center and its demonstrations, and the prompt states $c = 4$. We also add warnings of a 10% or 50% chance that the statement is wrong, which state the probability $1 - p$ that the description is irrelevant. Section 7.3 uses two designs for them. In the matched design, the warning says how the mean is distributed when the statement is wrong, and descriptions are in fact irrelevant at the stated rate. In an earlier design, the warning says only that the mean is then "unknown," and every description remains relevant. We also evaluate irrelevant descriptions without a warning (Section 7.5.4).

Table 7.1: **Evaluated model configurations.** "Full run" denotes the 2,000 tasks evaluated without reasoning, and "200 tasks" the first 200 of them; the open-weight results without reasoning on these 200 tasks are taken from the full run. Effort is the provider's reasoning-effort setting; "on" marks reasoning without such a setting. GPT-6 Sol at low effort and Haiku 5.5 at medium effort also run the matched design (Section 7.3) and the semantic task, and GPT-6 Sol runs both without reasoning as well; Haiku 5.5 also runs the wording experiment with reasoning. Section 7.5.5 gives the serving and decoding settings.

Model	Without reasoning	With reasoning	Run through
gpt-oss-20b	full run	200 tasks, low	Transformers; vLLM
Qwen3.5 9B	full run	200 tasks, on	Transformers; OpenRouter
Gemma 4 31B	full run	200 tasks, on	Transformers; Gemini API
Phi-4	full run	–	Transformers
OLMo 2 32B Instruct	full run	–	Transformers
Gemma 4 12B	full run	–	Transformers
Olmo 3 7B Think	–	200 tasks, on	vLLM
GPT-6 Sol	200 tasks	200 tasks, low	Codex command line
Claude Opus 4.6	200 tasks, high	200 tasks, high	Claude Code command line
Claude Haiku 5.5	–	200 tasks, medium and low	Claude Code command line

Table 7.1 lists the evaluated configurations. Using a stated precision requires weighting the description and the demonstrations by their precisions. Reasoning may help a model carry out this computation, so we select models that can answer with reasoning enabled and with it disabled under the same model identifier. They include open-weight and proprietary models from several developers. The main comparison

uses the same 200 tasks for five such models: gpt-oss-20b (OpenAI, 2025), Qwen3.5 9B (Qwen Team, 2026), Gemma 4 31B (Gemma Team, 2026), GPT-6 Sol (OpenAI, 2026), and Claude Opus 4.6 (Anthropic, 2026b). Two further models are evaluated only with reasoning. Claude Haiku 5.5 (Anthropic, 2026a) is another recent frontier model; its reasoning could not be disabled in our setup, and we run it at medium effort and, in a second run, at low effort. Olmo 3 7B Think (Team Olmo, 2025) is a small open-weight model that always reasons. To estimate behavior without reasoning more precisely and across more models, the *full run* evaluates six open-weight models without reasoning: gpt-oss-20b, Qwen3.5 9B, Gemma 4 31B, Phi-4 (14B; Abdin et al., 2024), OLMo 2 32B Instruct (Team OLMo et al., 2025), and Gemma 4 12B. It uses 2,000 tasks, the first 200 of which are the tasks of the main comparison. "Reasoning enabled" denotes the configured reasoning mode; individual replies may still contain no recorded reasoning output. Paired runs use the same model identifier, although serving and decoding settings differ for some models. Section 7.5 gives the evaluation details and the further results.

Bayesian reference. Our *prompt-based Bayesian reference* uses the observation noise, description uncertainty, and chance of irrelevance stated in the prompt, even when they differ from the generating values. Agreement with it therefore measures how a model uses the prompt's statistical information; it need not imply optimal prediction under the evaluation distribution. Without a description, the prompt says only that the mean is "unknown": a stated distribution for it would itself be a description, and the baseline should reflect only what a model already knows (Chapter 3). A warning in the earlier design likewise says only that the mean is then "unknown." In both cases, the reference therefore needs a baseline prior of its own. It takes a uniform one over the integers 100 to 999, the range of every number shown in a prompt, so its prediction without a description is approximately the demonstration mean. When the prompt asserts relevance, its prediction is approximately the posterior mean of the conjugate normal model of Section 2.5 (Neuenschwander et al., 2020),

$$\hat{w}_{\text{ref}} = \frac{c\,m + n\bar{y}}{c + n}, \tag{7.1}$$

where $\bar{y}$ is the demonstration mean; rounding and boundary effects on this expression are negligible. The description is equivalent to c additional demonstrations at m. In this idealized continuous model with a flat baseline prior, the squared error is

therefore $\sigma^2/(c+n)$ with the description and σ^2/n without it, so a correctly specified description has exact ESS c. When the prompt states a chance of irrelevance, the reference takes it as the prior probability of irrelevance and updates that probability from the demonstrations (Section 7.3).

We compute ESS against each model's own baseline curve and report the median over $n \in \{1, 2, 4\}$, which we call *summary ESS*; models rarely answer with a number at $n = 0$. Applied to the reference, this estimator departs from the idealized value because of finite-task sampling and interpolation: for the $c = 36$ description, it gives 34.6 on the first 200 tasks and the idealized value on the full run, and its values under a stated chance of irrelevance vary more between task sets (Section 7.5.2). Each model is therefore compared with the reference on the same tasks. We also regress predictions on m, $\bar{y}$, and an intercept. The coefficient on m is the *fitted description weight*; the reference weight is $c/(c+n)$ when relevance is asserted.

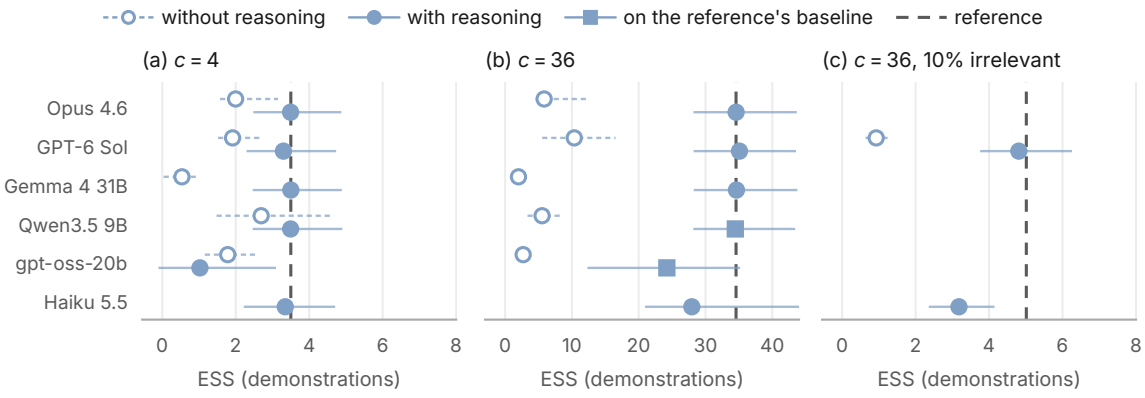


Figure 7.1: **With reasoning, language models use the precision and the reliability that a prompt states.** Summary ESS on 200 tasks at $\sigma = 60$, with 95% task-bootstrap intervals. Open points indicate reasoning disabled; filled points indicate reasoning enabled; vertical dashed lines show the prompt-based Bayesian reference. (a) Descriptions generated and stated at $c = 4$. (b) Descriptions generated and stated at $c = 36$. (c) For the two models run in this design, the $c = 36$ descriptions of (b) when the prompt states a 10% chance that the description is irrelevant; each task is evaluated with a relevant description and with an irrelevant one. Squares in (b) measure ESS on the reference's baseline curve, not the model's own, and so are not comparable with the other points. Intervals are clipped to the display.

7.2 Reasoning and Stated Precision

Precise descriptions are underweighted without reasoning. For $c = 36$, summary ESS ranges from 2.0 to 10.3 across the five models without reasoning, compared with 34.6 for the reference (Figure 7.1b). The shortfall grows with precision: at $c = 4$, the models reach 0.5 to 2.7, against 3.5 for the reference (Figure 7.1a), and at $c = 1$ they reach -0.2 to 0.9, against 1.0 (Table 7.6). The models also differ in how much they respond to the stated precision itself. When the prompt states $c = 4$ for the $c = 36$ tasks, reference ESS falls from 34.6 to 13.7. It stays above the 3.5 of the $c = 4$ tasks because the center is as accurate as before. Under the same change, GPT-6 Sol's ESS falls from 10.3 to 2.4, and the ESS of Opus 4.6 and of Gemma 4 31B falls by about half; Qwen3.5 9B and gpt-oss-20b change less (Table 7.6). Fitted weights fall short of the reference as well. With four demonstrations, the reference weights are 0.50 at stated $c = 4$ and 0.90 at stated $c = 36$; GPT-6 Sol's are 0.19 and 0.47, and Opus 4.6's are 0.20 and 0.40.

The full run shows the same pattern for six open-weight models on 2,000 tasks (Section 7.5.3): no model's summary ESS for the $c = 36$ description exceeds 7.6, and the weights of four of the six barely change with c. The six models' ESS rises with c mainly because a more precise description's center lies closer to the task. Rewording the description changes how much weight these models give it, but no tested wording yields the reference's sensitivity to precision.

Reasoning improves the use of stated precision. With reasoning enabled, Opus 4.6, GPT-6 Sol, and Gemma 4 31B attain approximately the reference ESS at both precisions (Figure 7.1a,b), and their ESS falls with the reference's when only the stated precision is lowered (Table 7.6). Haiku 5.5 at medium effort matches the reference at $c = 4$ and falls short at $c = 36$, with summary ESS 28 against 34.6. The shortfall is largest at $n = 1$, where its ESS is only 3.4 and more than a quarter of its replies contain no recorded reasoning output. At $n = 2$ its ESS matches the reference, and at $n = 4$ it is 28 against 33. The individual replies of these four models follow the reference. A reply implies a description weight, $(\hat{w} - \bar{y})/(m - \bar{y})$. This ratio is well determined on tasks with $|m - \bar{y}| \geq 30$; on those tasks at $n = 4$, between 89% and 100% of each model's replies have a weight within 0.05 of the reference's (Section 7.5.6). Enabling reasoning does not ensure agreement with the reference in every model,

however. Olmo 3 7B Think, not shown in Figure 7.1, averages demonstrations well but reaches summary ESS only 6.2 at $c = 36$ (Table 7.6); Section 7.5.6 examines how it combines the description with the demonstrations.

Qwen3.5 9B and gpt-oss-20b also approach the reference weights. Their own baseline curves flatten at larger n, and gpt-oss-20b makes substantial averaging errors. Some ESS crossings therefore fall beyond the grid. For these, the squares in Figure 7.1b use the reference's baseline curve instead: they report $n^*_{\text{ref}} - n$, where n^*_{ref} is the demonstration count at which that curve reaches the model's error. This comparison uses another predictor's baseline, so it is not the model's own ESS (Section 7.5.5). The remaining points use each model's own baseline.

The gains from reasoning also depend on how the description states its uncertainty. The informal wordings "give or take" and "±" do not define a prior standard deviation, and under them Haiku 5.5 often repeats m even with reasoning (Section 7.5.7).

7.3 Stated Reliability

Chapter 5 shows that uncertain relevance sharply lowers a description's value, and that demonstrations then carry evidence about relevance. To test both, we make the $c = 36$ description unreliable and say so. The prompt adds that there is a 10% chance that the statement is wrong, in which case the mean "was instead drawn uniformly at random from the whole numbers 100 to 999." The stated chance is also the true one. Each task is run twice: once with its relevant description, and once with an irrelevant one, whose center is drawn independently of the task. We average the two errors with weights 0.9 and 0.1, and call this the *matched design*. To limit cost and time, it covers two models, GPT-6 Sol and Haiku 5.5. We also run it with a 50% chance.

For the reference, a one-in-ten chance of irrelevance reduces ESS from 34.6 to 5.0 (Figure 7.1c), just as it reduces the Bayes-optimal ESS from 117 to 13.5 in Chapter 5. Most of what remains depends on checking the description against the demonstrations. Consider *fixed-weight averaging*, a predictor that keeps the chance of irrelevance at the stated 10%, whatever the demonstrations show. Its ESS is only 0.3, because it cannot discount a description that the demonstrations contradict. With reasoning, GPT-6 Sol reaches 4.8, and its replies are within 2 of the reference's prediction in 89% of cases with a relevant description and 92% with an irrelevant one. Haiku 5.5 reaches 3.2; 85% of its replies are within 2 of the reference with a relevant description and

60% with an irrelevant one. Without reasoning, GPT-6 Sol's ESS is 0.9, and at most 19% of its replies are within 2 of the reference. At a 50% chance, the reference's ESS is 0.9, and a predictor that keeps the chance fixed or ignores the warning does worse with the description than without it; GPT-6 Sol with reasoning is again within 0.2 of the reference (Table 7.2).

Table 7.2: **Unreliable descriptions with the stated chance of irrelevance.** Summary ESS in the matched design, with 95% task-bootstrap intervals, and the share of replies within 2 of the reference's prediction when the description is relevant (rel.) and irrelevant (irr.), pooled over $n = 1$, 2, and 4. With stated chance $1 - p$, the error with the description is p times the mean squared error with the relevant description plus $1-p$ times that with the irrelevant one. A value of ≤ -1.0 is censored: at $n = 2$ the description does worse than one demonstration alone.

	10% irrelevant			50% irrelevant		
	ESS	rel.	irr.	ESS	rel.	irr.
Reference	5.0 [3.9, 6.2]			0.9 [0.5, 1.2]		
Fixed weights	0.3 [−0.2, 1.0]			−0.7 [−1.0, −0.4]		
Ignores the warning	−0.1 [−0.6, 0.6]			≤ -1.0		
GPT-6 Sol, low effort	4.8 [3.8, 6.3]	89%	92%	0.8 [0.5, 1.2]	90%	92%
Haiku 5.5, medium effort	3.2 [2.4, 4.1]	85%	60%	0.6 [0.3, 0.9]	58%	52%
GPT-6 Sol, without reasoning	0.9 [0.6, 1.2]	19%	12%	−0.1 [−0.2, 0.1]	10%	12%

A model can revise the stated chance in light of the demonstrations only if the prompt says how the mean is distributed when the description is irrelevant. An earlier design shows what models do when it does not: the warning said only that the mean was then "unknown," and every description remained relevant. There, a stated 50% chance of irrelevance reduces reference ESS from 34.6 to 5.7 under the uniform baseline prior. With reasoning, the ESS of five of the six models in Figure 7.1 falls with the reference's, to between 3.8 and 4.2. The sixth, gpt-oss-20b, weights the description almost as if there were no warning. Its ESS is higher as a result, but only because the descriptions in this design are in fact relevant. Without reasoning, ESS falls by 36 to 62% for four models, against 84% for the reference; GPT-6 Sol's falls from 10.3 to 0.9. In the full run, no model's ESS falls in proportion to the stated probability (Section 7.5.3).

The five models' ESS is below the reference's because the reference also updates the

stated chance on the demonstrations: demonstrations close to the stated center make relevance more probable, and distant ones less. The models apply the stated chance as given. They closely match fixed-weight averaging. Under it, each component's prediction conditions on the demonstrations, while the components keep their prior weights, p and $1 - p$. In the scalar approximation, the resulting weight on m is $pc/(c + n)$ (Table 7.7). Fixed-weight averaging has ESS 4.4, and 80 to 100% of the replies of Opus 4.6, GPT-6 Sol, Gemma 4 31B, and Qwen3.5 9B are within 2 of its prediction (Section 7.5.6). The gap to the reference widens with demonstrations. At $n = 4$, the reference's ESS reaches 16.5; fixed-weight averaging reaches 8.6, and the five models reach between 3.8 and 8.6.

Updating the chance is Bayesian model averaging, which uses posterior component probabilities. These depend on both component marginal likelihoods and hence on the baseline prior under irrelevance. The original warning leaves the baseline prior unspecified, saying only that the mean is otherwise "unknown," so the prompt does not determine a unique posterior relevance probability: a model must supply the missing prior from its pretrained knowledge, and the reference uses its uniform baseline prior. The relevant descriptions of the matched design supply the comparison. There, GPT-6 Sol and Haiku 5.5, in their main reasoning configurations, answer the same tasks with that prior stated explicitly.

With the baseline prior stated and a 50% probability of irrelevance, both models update the chance as the reference does. On tasks where the reference and fixed-weight predictions differ by at least 5, the share of replies closer to the reference rises from 0–5% to 92–97% for GPT-6 Sol and from 21–41% to 87–95% for Haiku 5.5, where each range is over $n = 1$, 2, and 4. The share of all replies within 2 of the reference prediction rises from 10% to 90% for GPT-6 Sol and from 23% to 58% for Haiku 5.5. GPT-6 Sol's ESS rises from 4.2 to 5.4. Haiku 5.5's rises less, from 3.8 to 4.2, because a minority of its replies lie far from the reference (Section 7.5.6).

Replies could move closer to the reference merely by giving the description a larger constant weight, because the reference's posterior relevance probability exceeds the stated 50% on most tasks. To test whether the weights instead vary with the evidence, we divide each reply's description weight by $c/(c + n)$, which gives its *normalized description weight*. Under the scalar approximation in Equation (7.1), the reference's normalized weight equals its posterior relevance probability, π_n, whereas fixed-weight averaging keeps it constant at p. Regressing the models' normalized weights on π_n gives

slopes 0.96 [0.84, 1.04] and 0.98 [0.82, 1.15] for GPT-6 Sol and Haiku 5.5, respectively, compared with 0.01 [−0.09, 0.11] and 0.22 [−0.01, 0.45] under the original warning (Section 7.5.6).

At a stated 10% chance of irrelevance, the two weightings rarely differ by much, so this comparison is uninformative. In that condition, stating the prior lowers Haiku 5.5's ESS from 32.6 to 15.1, against 27.5 for the reference, because a minority of its replies lie far from the reference prediction (Section 7.5.6).

Apart from the matched design, the experiments above keep descriptions relevant and vary only what the prompt states. The full run and the main comparison also include irrelevant descriptions that the prompt presents as relevant (Section 7.5.4; Table 7.6). In the full run, the models give an irrelevant description less weight than a relevant one, and less than the reference, which treats it as relevant.

7.4 A Semantic Task

The numerical descriptions state a distribution over the task, whereas descriptions in practice usually state constraints in ordinary language that the model must interpret. We therefore design a message-filing task that is closer to such descriptions and still admits an exact Bayes-optimal predictor. As in Bayesian models of rule-based concept learning (Goodman et al., 2008), a hidden filing policy is one of a finite set of rules, here 60 under a uniform prior. A message has six binary properties, and a policy files it into folder A when two of them take required values. Models give the probability of A, and we score it by squared error. Policy notes constrain the policy in ordinary language: a broad note leaves 20 compatible policies, and a specific note leaves 2 (Section 7.5.8). We evaluate the two models of the matched design, GPT-6 Sol and Haiku 5.5.

With reasoning, Haiku 5.5 and GPT-6 Sol obtain ESS 9.9 and 9.5 for the specific note at $n = 2$, compared with 11.4 for the reference. For unreliable notes, the generator draws the note independently of the task with probability 0.5, and the prompt says that a wrong note says nothing about the policy. We compare Bayesian predictions with fixed-weight averaging, which in this task keeps the stated weight on the note unless the demonstrations rule the note out. Among queries where these predictions differ by at least 0.05, 77–99% of the two models' predictions are closer to the Bayesian reference. The agreement with posterior weighting therefore extends to

ordinary-language constraints within this finite task family.

7.5 Additional Results and Evaluation Details

This section gives the evaluation details and the full results that Sections 7.2–7.4 summarize. Sections 7.5.1 and 7.5.2 give the prompts, readouts, and ESS estimation, and Sections 7.5.3 and 7.5.4 report the full run; unless stated otherwise, these four sections use the full run, which Appendix A reports at every evaluated demonstration count. Sections 7.5.5–7.5.7 cover the reasoning runs on the first 200 tasks, and Section 7.5.8 the semantic task, which uses a separate set of 200 tasks.

7.5.1 Prompts, Tasks, and Model Readouts

Prompt. The following text is supplied as the user turn in the model's chat template:

```
The numbers below are drawn from a normal distribution with standard deviation 60.
Its mean was drawn from a normal distribution with mean 510 and standard deviation 30.
548
431
Predict the next number.  Reply with the number only.
```

Without a description, the second sentence is "Its mean is unknown." To state unreliability, a third sentence is added: "There is a 10% chance that this statement is wrong, in which case the mean is unknown." In the rerun that states the baseline prior, the final clause becomes "in which case the mean was instead drawn uniformly at random from the whole numbers 100 to 999." With no demonstrations, the final question is "Predict the first number."

Task generation. A relevant description value is sampled conditional on the task so that the joint distribution satisfies $m \sim \mathcal{N}(500, 100^2 - \tau^2)$ and $w \mid m \sim \mathcal{N}(m, \tau^2)$. A task is redrawn if a displayed number or description value falls outside 100 to 999; this occurs for less than 1% of tasks. The full run also includes $\sigma = 30$ conditions. Both standard deviations stated in the prompt are then halved, preserving $c \in \{1, 4, 36\}$. The qualitative pattern is unchanged (Table A.1).

Execution. The full run uses Hugging Face Transformers at each model's native precision, with checkpoints `openai/gpt-oss-20b`, `microsoft/phi-4`, `allenai/`

`OLMo-2-0325-32B-Instruct`, `Qwen/Qwen3.5-9B`, `google/gemma-4-12B-it`, and `google/gemma-4-31B-it`. Tasks are sampled by a seeded generator. Reasoning is disabled where the chat template supports this option. For gpt-oss-20b, generation begins in the final channel, after the channel normally used for reasoning.

Distribution and written readouts. For gpt-oss-20b, Phi-4, and OLMo 2 32B without reasoning, the prediction is the mean of the first-token distribution, renormalized over the 900 three-digit number tokens. We call this the *distribution readout*. Other models, and all reasoning runs, use the number in the final answer, called the *written readout*. We also evaluate written readouts for the three distribution-readout models as a robustness check (Table 7.3 for gpt-oss-20b, and Table 7.5). OLMo 2 32B assigns 41% of first-token probability to numbers without a description at $n = 1$, and 75% at $n = 2$. Its distribution readout at $n = 1$ therefore summarizes a minority of the first-token probability mass.

Table 7.3: **Description ESS and weights in the full run without reasoning.** Left: median ESS over $n = 1$, 2, and 4, for relevant descriptions with $c = 1$, 4, and 36, and for the $c = 36$ description when irrelevant or with a stated 10% or 50% chance of irrelevance ("10% wrong" and "50% wrong", after the prompt's wording). Brackets give 95% bootstrap intervals. The censoring marker $\leq$ indicates that one demonstration alone already outperforms the description-conditioned prediction at a boundary used in the summary. Right: the fitted description weight at $n = 1$; the Bayesian reference weight is $c/(c + n)$. All results use $\sigma = 60$. "Written" denotes the written readout. Reference values are computed on the same 2,000 tasks and differ from those on the first 200 tasks (Section 7.5.2).

	ESS, median over $n = 1$, 2, 4						weight, $n = 1$		
	relevant			$c = 36$			relevant		
Model	$c = 1$	$c = 4$	$c = 36$	irrelevant	10% wrong	50% wrong	$c = 1$	$c = 4$	$c = 36$
Reference	1.3	4.4	36	≤ -1.0	22	5.8	0.50	0.80	0.97
gpt-oss-20b	0.9 [0.7, 1.0]	2.0 [1.7, 2.3]	2.8 [2.6, 3.1]	≤ -0.4	2.4 [1.9, 2.6]	2.3 [1.9, 2.7]	0.53	0.48	0.49
Phi-4	0.7 [0.4, 0.9]	4.1 [3.5, 4.6]	7.6 [6.5, 8.6]	≤ -1.0	8.7 [7.8, 9.8]	9.7 [8.7, 11]	0.82	0.76	0.74
OLMo 2 32B	1.9 [1.7, 2.1]	3.4 [3.2, 3.9]	5.3 [4.5, 6.3]	≤ -0.6	6.0 [5.6, 7.1]	5.4 [5.0, 6.0]	0.72	0.70	0.70
gpt-oss-20b (written)	0.2 [0.1, 0.4]	2.0 [1.9, 2.6]	3.7 [3.0, 4.4]	≤ -1.0	2.4 [2.1, 2.6]	2.3 [2.1, 2.6]	0.64	0.52	0.52
Qwen3.5 9B (written)	0.2 [−0.1, 0.4]	2.7 [2.1, 3.2]	4.0 [3.5, 4.5]	≤ -1.0	0.6 [0.3, 1.0]	1.0 [0.6, 1.5]	0.81	0.76	0.80
Gemma 4 12B (written)	0.0 [−0.1, 0.1]	0.5 [0.3, 0.7]	1.8 [1.4, 2.2]	≤ -0.7	1.3 [1.1, 1.6]	2.7 [2.5, 3.1]	0.34	0.24	0.59
Gemma 4 31B (written)	0.0 [0.0, 0.0]	0.2 [0.1, 0.3]	1.4 [1.2, 1.6]	≤ -0.7	0.6 [0.5, 0.7]	0.5 [0.4, 0.6]	0.48	0.49	0.72

Numeric response rates. For the distribution readout, the numeric response rate is the probability mass assigned to number tokens at the first position. For the written readout, it is the fraction of replies containing a three-digit number. Tables 7.5, A.1, A.2, and A.3 report this quantity as "number". Tables 7.5 and A.3 italicize cells whose rate is below three quarters.

Model inclusion. ESS requires a baseline curve built from numeric answers, so we retain a model only if it meets two conditions. It must produce a number in at least three quarters of its replies overall. It must also do so in at least half of its replies without a description, averaged over $n = 1$, 2, and 4 and both noise levels, because these replies determine the baseline curve used for summary ESS. This criterion excludes MiniCPM5 2B (MiniCPM Team, 2026), OLMo 2 7B Instruct, and OLMo 2 13B Instruct, whose replies begin with a number in 62%, 20%, and 10% of cases, respectively. These models often begin with explanations such as "To predict…" or "Given…" despite the numeric-only instruction. For example, OLMo 2 7B's summary ESS for the $c = 36$ description is 21, but this value comes from a readout that covers only a minority of the model's first-token probability mass.

Written-readout implementation and checks. The written readout extracts the number in the last `\boxed{}` expression in the final answer, if present. Otherwise, it takes the last number with three digits before an optional decimal part, retaining the decimals. A digit group after a thousands separator, as in 14,945, does not count as a qualifying number. In the full run, decoding is greedy with a limit of eight new tokens, so the reply is the number itself. Those predictions were recorded using an earlier readout that took the first qualifying number and retained only its integer part. Replies without a qualifying number are excluded from both error curves and weight regressions; the reported response rates identify the extent of this exclusion.

For gpt-oss-20b, written-readout weights differ from distribution-readout weights by at most 0.21 across all conditions and demonstration counts. For $c = 1$, 4, and 36, written-readout summary ESS is 0.2, 2.0, and 3.7, compared with 0.9, 2.0, and 2.8 from the distribution readout. A single greedy response is a noisier summary than the distribution mean. Phi-4 produces a number in only 58% of written replies across all conditions, so Table 7.3 reports only its distribution readout. In the wording experiment, which uses written replies, Phi-4's numeric response rate nevertheless

exceeds three quarters for the original wording. OLMo 2 32B produces a number in only 5%, 36%, and 48% of written replies without a description at $n = 1$, 2, and 4, respectively, with $\sigma = 60$; we therefore do not report its written-readout ESS. Its written-readout weights for the three relevant descriptions at $n = 1$ are 0.83, 0.81, and 0.83, compared with distribution-readout weights of 0.72, 0.70, and 0.70.

7.5.2 ESS Estimation and the Bayesian Reference

Measurement. We apply Definition 3.1 to squared error against the hidden mean. In the full run, the baseline curve is evaluated at $n = 1, 2, 4, \ldots, 64$. We interpolate squared error, rather than its square root, linearly in $\log n$. Let t be the first interpolated demonstration count at which the baseline curve matches the description-conditioned error; the ESS at n is $t - n$. Figure 7.2 shows the full run's curves.

If the description-conditioned error lies below the entire baseline curve up to $t_{\max}$, we report $\mathrm{ESS}_n > t_{\max} - n$. Because the curve starts at one demonstration, we use the marker $\mathrm{ESS}_n \leq 1 - n$ when one demonstration alone already gives lower error. These markers record censoring at the evaluation boundaries; they do not define an extrapolated matching count or an exact ESS when the crossing does not exist. Summary ESS excludes $n = 0$ because models rarely return a number without a description or demonstrations. At large n, ESS is strongly affected by slow baseline learning and excessive weight on the $c = 1$ and $c = 4$ descriptions. At $\sigma = 60$ and $n = 64$, model root-mean-square error without a description ranges from 9 to 20, compared with 7.5 for the reference. A learner with a flatter baseline curve can assign a larger ESS to the same error reduction. Description-weight estimates complement ESS because they do not use that curve.

Bootstrap replicates resample tasks jointly for the description-conditioned and baseline curves and recompute the complete ESS statistic. The intervals therefore measure variation across tasks; they do not include interpolation bias, quantified below. We use 500 replicates with a fixed seed and the same resampled tasks for every model.

Reference prior and exact ESS. Integer rounding, finite-range truncation, and the task-rejection rule are not part of the exact ESS c of Section 7.1, so we compute the reported reference values on the same generated tasks as the models.

Applying the empirical ESS procedure to the full-run reference gives 1.3, 4.4, and 36 for $c = 1$, 4, and 36 at $\sigma = 60$, and 1.2, 4.7, and 43 at $\sigma = 30$. Interpolation in $\log n$ induces an upward bias, while finite-task sampling can shift the estimate in either direction. We use the same estimator for the reference and the models. On the first 200 tasks, the grid additionally includes $n = 40$ and 48 near the $c = 36$ crossing, and the reference values are 1.0, 3.5, and 34.6. With a stated chance of irrelevance, the reference's summary ESS varies more across task samples: on 20,000 tasks it is 7.3 at a stated 50% and 23.4 at 10%, compared with 5.8 and 22 on the full run's 2,000 tasks and 5.7 and 27.5 on its first 200.

The actual task prior, $\mathcal{N}(500, 100^2)$, is not stated in the prompt. A reference using that prior would have baseline precision equivalent to 0.36 demonstrations at $\sigma = 60$ and 0.09 at $\sigma = 30$. At $\sigma = 60$, the corresponding idealized continuous Gaussian model would give a relevant description exact ESS $c - 0.36$, namely 0.64, 3.64, and 35.64. This is the scalar analogue of $c - a_0^{-1}$ in Chapter 4. We use the prompt-based reference because the models are not given the actual task prior. This distinction matters under misspecification: unlike the correctly specified Bayesian predictor in Chapters 4–5, a reference that incorrectly assumes an irrelevant description to be relevant can have negative ESS, indicating increased prediction error (Reimherr et al., 2021).

Sensitivity to interpolation. The full-run grid doubles n, so interpolation can affect ESS. The reference's idealized errors are σ^2/n without a description and $\sigma^2/(c+n)$ with one. On these errors, interpolation in $\log n$ overestimates ESS by 0.2, 0.3, and 1.8 for $c = 1$, 4, and 36. Linear interpolation in n gives overestimates of 0.3, 0.6, and 4.1.

For the models without reasoning, the relevant crossings are unique. Among the twelve baseline curves across the two noise levels, only Phi-4 at $\sigma = 60$ increases, slightly, from 20.3 at $n = 32$ to 20.4 at $n = 64$. Table 7.4 compares four estimators: interpolation in $\log n$; linear interpolation in n; a least-squares non-increasing fit using the pool-adjacent-violators algorithm; and a relative-least-squares fit of $A/(n+n_0)+B$ to the baseline errors. The fourth estimator's curve, $A/(n + n_0) + B$, is a Bayesian learning curve plus an error floor, and it recovers exact ESS when fit to the reference's idealized errors.

Across these choices, no model's summary ESS changes by more than 1.4 demonstrations. For the $c = 36$ description, no model exceeds 8.1, while the reference ranges

from 35 to 38. Differences become larger at higher demonstration counts, where a flat learning curve turns a small error difference into a large change in ESS. For example, gpt-oss-20b at $n = 8$ and $c = 36$ has ESS 17 under the estimator in Table A.1, but 9.2 under the fitted curve.

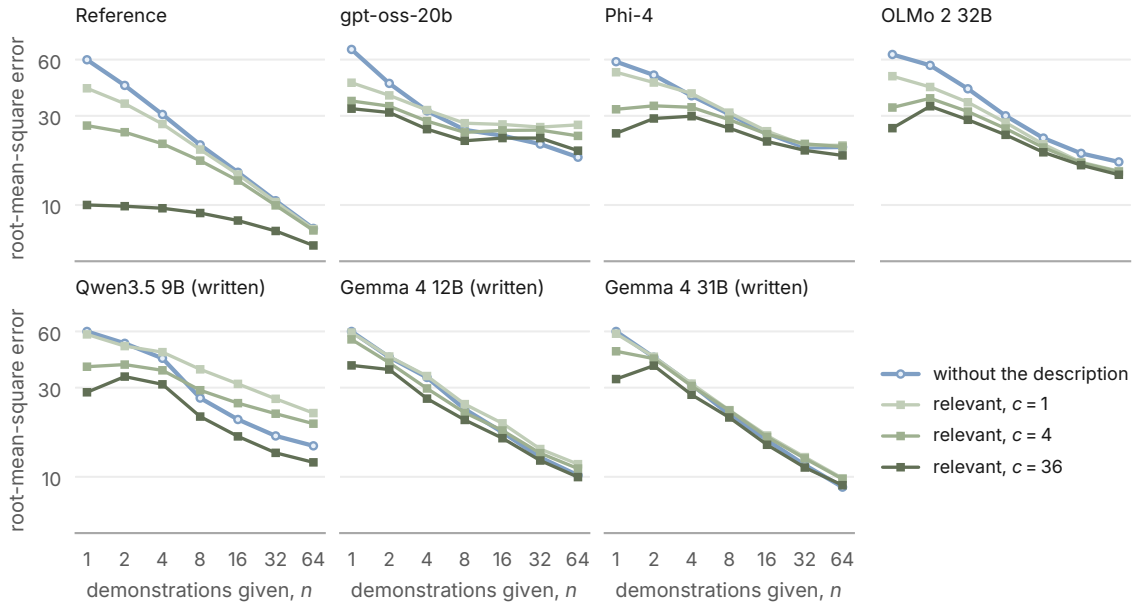


Figure 7.2: **Learning curves used to estimate language-model ESS.** Root-mean-square error versus demonstration count for the reference and six models in the full run without reasoning, at $\sigma = 60$. Blue curves omit the description; green curves use relevant descriptions, with darker shades indicating higher c. Replies without a number are excluded. ESS is the horizontal distance from the description-conditioned point at n to the baseline curve at the same error. For four models, error with $c = 36$ increases between one and two demonstrations because the models shift toward the observations more than the reference does (Table A.3).

7.5.3 Results of the Full Run

Summary ESS. Across the six models, the $c = 36$ description has summary ESS 1.4 to 7.6, and its ESS is at most 15 at any of $n = 1$, 2, and 4; the reference's ESS on these tasks is 36 (Table 7.3). The $c = 1$ and $c = 4$ descriptions have summary ESS 0.0 to 1.9 and 0.2 to 4.1, compared with 1.3 and 4.4. Summary ESS thus rises with c. The

Table 7.4: **Sensitivity of summary ESS to baseline-curve estimation.** Median ESS over $n = 1$, 2, and 4 for relevant descriptions, using the full run without reasoning at $\sigma = 60$. "Log n" is the primary estimator; "linear" interpolates in n; "monotone" applies a non-increasing fit; and "fit" uses $A/(n + n_0) + B$. The reference's idealized ESS values are 1, 4, and 36.

	$c = 1$				$c = 4$				$c = 36$			
Predictor	log n	linear	monotone	fit	log n	linear	monotone	fit	log n	linear	monotone	fit
Reference	1.3	1.4	1.3	1.0	4.4	4.7	4.4	4.0	36	38	36	35
gpt-oss-20b	0.9	1.0	0.9	0.9	2.0	2.4	2.0	2.4	2.8	2.9	2.8	3.4
Phi-4	0.7	0.8	0.7	0.4	4.1	4.4	4.1	3.6	7.6	8.1	7.6	7.1
OLMo 2 32B	1.9	2.1	1.9	1.4	3.4	3.7	3.4	3.3	5.3	5.8	5.3	5.5
Qwen3.5 9B (written)	0.2	0.3	0.2	−0.1	2.7	2.9	2.7	1.3	4.0	4.3	4.0	2.8
Gemma 4 12B (written)	0.0	0.0	0.0	0.0	0.5	0.6	0.5	0.4	1.8	1.9	1.8	1.6
Gemma 4 31B (written)	0.0	0.0	0.0	−0.1	0.2	0.3	0.2	0.0	1.4	1.7	1.4	1.0

description weights below indicate that the rise comes mainly from the description value, which lies closer to the task at higher c, rather than from the stated precision.

Description weights. At $\sigma = 60$, with no demonstrations and no stated reliability, the models return the description value (weight 0.92 to 1.10), although Phi-4 and OLMo 2 32B then begin only 20 to 23% and 50 to 59% of their replies with a number. At $n = 1$, the weight of gpt-oss-20b, Phi-4, and OLMo 2 32B changes by at most 0.08 from $c = 1$ to $c = 36$, and if anything falls, whereas the reference weight rises from 0.50 to 0.80 to 0.97 (Table 7.3). At every n from 1 to 64 and at both noise levels, the spread of these three models' weights across c is at most 0.16, compared with a reference spread of 0.34 to 0.71 (Table A.3). The weights of the two Gemma models do rise with c, by about 0.25 between $c = 1$ and $c = 36$ at $n = 1$, half the reference rise. The weight of Qwen3.5 9B does not. Models therefore underweight a precise description and, with many demonstrations, overweight the least precise one. At $n = 64$, gpt-oss-20b still assigns weight 0.23 to the $c = 1$ value, compared with reference weight 0.02, and that description raises its error (ESS −57).

Stated unreliability. A stated 10% or 50% chance of irrelevance barely changes the ESS of gpt-oss-20b and OLMo 2 32B, and Phi-4's ESS even rises, from 7.6 to 8.7 and 9.7. Qwen3.5 9B and Gemma 4 31B have lower ESS, but by about as much for 50% as for 10%. Gemma 4 12B's ESS falls at 10% and rises at 50% (Table 7.3). No

model's ESS falls in proportion to the stated probability, whereas reference ESS falls to 22 and 5.8.

7.5.4 Irrelevant Descriptions and Wording

Apart from the matched design, the experiments above keep every description relevant, and all of them state its precision in one wording. This section relaxes each of these choices in the full run.

Irrelevant descriptions. An irrelevant description uses an independently drawn value. At $c = 36$, these descriptions have negative ESS for every model and for the prompt-based reference, which treats them as relevant. The models assign less weight to the description than the reference does and consequently incur lower error in this condition: at $n = 4$, their root-mean-square errors range from 35 to 64, compared with 124 for the reference. They also discount it relative to a relevant description: at $n = 4$, their weight on an irrelevant $c = 36$ value is 0.01 to 0.33, compared with 0.18 to 0.44 on a relevant one.

Wording without reasoning. Descriptions in practice express uncertainty in many ways. To test whether the results depend on the original wording, we repeat the three relevant-description conditions and the no-description baseline at $n = 1$, 2, and 4 using three alternative description sentences: "Its mean is about m, give or take τ.", "Mean: $m \pm \tau$", and "An earlier estimate put its mean at m, with a standard error of τ." Table 7.5 uses the full-run tasks and written readouts. In that table, a fitted coefficient above 1 means that a unit change in the description value corresponds to more than a unit change in the prediction, with the demonstration mean held fixed in the regression.

Wording substantially changes the weight assigned to the description. For Gemma 4 12B with $c = 1$ and $n = 1$, this weight is 0.34 under the original wording and 1.00 under each alternative. For gpt-oss-20b, it ranges from 0.17 to 0.67. Nevertheless, none of the tested wordings yields the reference's sensitivity to prior precision: across models and wordings, the weight difference between $c = 36$ and $c = 1$ is at most 0.25 at $n = 1$ and 0.44 at $n = 4$, compared with reference differences of 0.47 and 0.70. The informal descriptions do not uniquely specify a probability distribution, so this

comparison also varies how precisely the task is specified.

Table 7.5: **Description weights under four wordings without reasoning.** Weights are reported for the three relevant descriptions at $n = 1$ and 4, using written readouts on the full-run tasks. "Number" denotes the numeric response rate averaged over demonstration counts; a weight is italicized when the response rate at that count is below three quarters. The reference weights are 0.50, 0.80, and 0.97 at $n = 1$, and 0.20, 0.50, and 0.90 at $n = 4$. Wording changes overall reliance on the description, but the observed change with stated precision remains smaller than the reference change.

			weight, $n = 1$			weight, $n = 4$		
Model	Wording of the description	number	$c = 1$	$c = 4$	$c = 36$	$c = 1$	$c = 4$	$c = 36$
Gemma 4 12B	normal, mean m, sd τ (ours)	1.00	0.34	0.24	0.59	0.19	0.24	0.31
	about m, give or take τ	1.00	1.00	1.00	1.00	0.87	0.76	0.82
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.60	0.66	0.74
	earlier estimate m, s.e. τ	1.00	1.00	1.00	1.00	0.45	0.54	0.67
Gemma 4 31B	normal, mean m, sd τ (ours)	1.00	0.48	0.49	0.72	0.02	0.04	0.18
	about m, give or take τ	1.00	1.00	1.00	1.00	1.01	1.22	1.45
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.90	1.03	1.15
	earlier estimate m, s.e. τ	1.00	0.94	0.94	0.96	0.09	0.23	0.41
gpt-oss-20b	normal, mean m, sd τ (ours)	1.00	0.64	0.52	0.52	0.23	0.29	0.34
	about m, give or take τ	1.00	0.24	0.17	0.38	0.28	0.29	0.38
	Mean: $m \pm \tau$	1.00	0.17	0.26	0.40	0.23	0.24	0.30
	earlier estimate m, s.e. τ	1.00	0.67	0.63	0.67	0.28	0.37	0.42
OLMo 2 32B	normal, mean m, sd τ (ours)	0.61	0.83	0.81	0.83	*0.20*	*0.29*	0.38
	about m, give or take τ	0.82	0.93	0.92	0.96	0.60	0.57	0.65
	Mean: $m \pm \tau$	0.82	0.88	0.83	0.88	0.44	0.46	0.55
	earlier estimate m, s.e. τ	0.82	0.95	0.94	0.95	0.49	0.57	0.65
Phi-4	normal, mean m, sd τ (ours)	0.77	0.96	0.78	0.76	*0.38*	*0.37*	*0.38*
	about m, give or take τ	0.97	1.02	1.01	1.02	0.65	0.57	0.62
	Mean: $m \pm \tau$	0.83	*0.99*	*0.87*	*0.87*	0.45	0.42	0.50
	earlier estimate m, s.e. τ	0.81	0.88	0.76	0.73	*0.47*	*0.49*	0.54
Qwen3.5 9B	normal, mean m, sd τ (ours)	1.00	0.81	0.76	0.80	0.26	0.27	0.44
	about m, give or take τ	1.00	0.99	1.00	0.99	0.35	0.43	0.57
	Mean: $m \pm \tau$	1.00	1.08	1.25	1.14	0.19	0.27	0.38
	earlier estimate m, s.e. τ	1.00	0.87	0.86	0.86	0.61	0.75	0.82

7.5.5 Reasoning Runs and Execution Settings

Comparisons between modes. As in Section 7.1, "with reasoning" and "without reasoning" refer to generation settings, and statements about individual replies refer to recorded reasoning output. Paired runs share the model identifier and prompts, but reasoning settings, serving software, readout, and retry policy vary across the runs; no

run has access to calculation tools. The comparisons therefore concern the evaluated configurations and do not identify a unique internal mechanism.

Shared evaluation prompts. Most configurations in Figure 7.1 are evaluated on the same 7,800 prompts from the first 200 tasks. These include no-description prompts at $n = 1$, 2, 4, ..., 64, with additional points at $n = 40$ and 48 near the $c = 36$ crossing, and every description condition at $n = 1$, 2, and 4. The gpt-oss-20b and Olmo 3 7B Think runs omit the two additional baseline points and contain 7,400 prompts. Results without reasoning for the open-weight models are taken from the full run restricted to these tasks. At least 99.7% of replies in each run contain a number.

Open-weight reasoning models. With reasoning, gpt-oss-20b runs in vLLM using greedy decoding, low reasoning effort, and at most 4,096 new tokens. It reaches the final channel in 99.9% of replies and uses 322 tokens on average, and at least 97% of replies reach it in every cell. An earlier run with a limit of 384 tokens finished only 28 to 54% of the replies in the stated-reliability conditions. That run seemed to show a difference between a stated 10% and a stated 50%, but the difference came from which replies finished. Qwen3.5 9B runs through OpenRouter at temperature 0, with reasoning enabled and a limit of 24,576 new tokens. We pin requests to DeepInfra, which serves the model in bf16, because quantization varies across providers. Of its replies, 2,305 repeat themselves until they reach the token limit, a failure its model card warns of. Each is rerun once with the model card's recommended reasoning sampling settings: temperature 1.0, top-p 0.95, top-k 20, and presence penalty 1.5. Eleven remain unfinished and one request fails.

Gemma 4 31B (`gemma-4-31b-it`) runs through the Gemini API at temperature 0. Reasoning is enabled by default, and the model uses 2,613 reasoning tokens on average, with the same 24,576-token limit. Of its replies, 435 reach the limit; all finish after one rerun with the model card's recommended sampling settings: temperature 1.0, top-p 0.95, and top-k 64.

Olmo 3 7B Think (`allenai/Olmo-3-7B-Think`) runs in vLLM with greedy decoding and at most 32,768 new tokens, the limit its model card recommends; at 8,192 tokens, only 12 of 20 pilot replies finished. In the main run, 169 replies reach the 32,768-token limit. We rerun each of them once, with temperature 0.6 and top-p 0.95; 17 remain unfinished. The model generates 9,215 tokens on average and writes a worked solution

before most answers. Its long greedy outputs are sensitive to hardware: an earlier stated-precision run on a different GPU produces the same final number on only half the prompts. We therefore report one run; Section 7.5.6 analyzes its prediction rules.

Models served through command-line agents. GPT-6 Sol (`gpt-6-sol`, the model identifier served by Codex during the runs) runs through the Codex command-line agent in a read-only sandbox, with an empty configuration directory so that no user instructions or settings load, and with every tool that can run code disabled. We evaluate the model with reasoning disabled (no reasoning tokens) and enabled at low effort.

Claude Opus 4.6 (`claude-opus-4-6`) and Haiku 5.5 (`claude-haiku-5-5`) run through the Claude Code command line, with one prompt per call, no tools, no project or user instructions, and the system prompt "You are a helpful assistant." Opus 4.6 uses high effort in both evaluated modes, with reasoning either enabled or disabled; the disabled mode emits no reasoning tokens. With reasoning enabled, at least 99% of replies in each description condition include reasoning output, with an average of 599 reasoning tokens across all replies. Without a description, only 1%, 24%, and 79% of replies include reasoning output at $n = 1$, 2, and 4, respectively. Without reasoning, Opus 4.6 still writes out intermediate steps in 64 of its 7,800 replies; accounting for these replies changes no summary ESS by more than 0.05. Haiku 5.5 is evaluated at medium and low effort. Both Claude models use the command line's default sampling settings, which the command line does not report, and we collect one reply per prompt.

At medium effort, Haiku 5.5 includes reasoning output in 37 to 88% of replies at $n = 1$ across conditions and in nearly all replies from $n = 2$ onward, with 270 reasoning tokens on average. Its no-description error stays close to the reference through $n = 48$; at $n = 16$, for example, it is 15.8 against 14.8, with two replies far from the sample mean. Its error rises at $n = 64$, where it divides the sum by 60 instead of 64 in 9 of 200 replies.

Run dates. The full run took place on 2–3 October 2026. Its control conditions and gpt-oss-20b's reasoning run followed on 20 October. Haiku 5.5 was evaluated on 20 October for the main runs and 4 November for wording, the relevant descriptions of the matched design, and the semantic task. Opus 4.6, Qwen3.5 9B, Gemma 4 31B,

and Olmo 3 7B Think were evaluated on 3–4 November. GPT-6 Sol was evaluated on 4 November. The irrelevant descriptions of the matched design, and GPT-6 Sol's run of that design without reasoning, were evaluated on 9 November. The GPT-6 Sol logs do not record a model version beyond the identifier above.

ESS when baseline arithmetic fails. With reasoning, gpt-oss-20b and Qwen3.5 9B nearly match the reference description weight: at $n = 4$ and $c = 36$, their fitted weights are 0.89 and 0.90, compared with 0.90 for the reference. Their errors without descriptions, however, stop falling at larger n. gpt-oss-20b makes errors when averaging long sequences: its no-description root-mean-square error is 19.0 at $n = 32$ and 24.7 at $n = 64$, compared with reference errors 10.0 and 7.4. At these counts, 37 and 62 replies, respectively, differ from the sample mean by more than 20. Qwen3.5 9B's error levels off near 11 from $n = 32$ onward (11.2 at $n = 64$); in each cell, at most eleven of its replies differ from the sample mean by more than 20.

For these models, some description-conditioned errors never cross the baseline curve within the evaluated grid, so Table 7.6 also reports, in parentheses, the comparison on the reference's baseline, $n^*_{\text{ref}} - n$, defined in Section 7.2. gpt-oss-20b also makes arithmetic errors with descriptions: about 2% of its replies are more than 40 from the reference prediction, from errors such as exchanging the two variances. These errors reduce gpt-oss-20b's ESS. When the prompt states a 10% chance of irrelevance, for example, its summary ESS is 1.3. On the reference's baseline, its $c = 36$ description is worth 24 demonstrations, well below the reference's 34.6, although its fitted weight matches the reference's. A fitted weight is insensitive to such errors, whereas ESS reflects them.

Opus 4.6 and GPT-6 Sol do not exhibit these arithmetic failures. Opus 4.6 has root-mean-square error 7.4 at $n = 64$, matching the reference on these tasks. For these two models, changing between the model and reference baselines changes ESS by at most 1.

7.5.6 Prediction Rules and Relevance Inference

Per-response description weights. At $n = 4$, we compute the per-response description weight $(\hat{w} - \bar{y})/(m - \bar{y})$. This quantity describes one response and differs from the fitted coefficient obtained by regressing predictions across tasks: a fitted

Table 7.6: **Description ESS across reasoning configurations.** Results use the first 200 tasks at $\sigma = 60$. Summary ESS uses each run's own baseline curve; parentheses report the value computed using the reference baseline where it differs. "Stated irrelevance" gives the stated probability that a description generated at $c = 36$ is irrelevant. The last column fixes the generating precision at 36 and states it as 4. Each run names the mode and, in parentheses, the provider's effort setting. An upper bound, $\leq$, indicates that one demonstration alone gives lower error; > 60 denotes a crossing beyond the grid. Reference values use the same tasks and estimator and therefore differ from those in Table 7.3 (Section 7.5.2). The open-weight rows without reasoning take these 200 tasks from the full run, whose results on all 2,000 tasks are in Table 7.3. The runs also include three further fixed-value conditions, not shown.

					Stated irrelevance		
Run	$c = 1$	$c = 4$	$c = 36$	Irrelevant	10%	50%	Stated $c = 4$ Generated $c = 36$
Reference	1.0	3.5	34.6	≤ -1.0	27.5	5.7	13.7
GPT-6 Sol, without reasoning	0.9	1.9	10	≤ -1.0	3.3	0.9	2.4
GPT-6 Sol, with reasoning (low)	0.9	3.3	35 (34)	≤ -1.0	30 (29)	4.2	11
Opus 4.6, without reasoning	0.9	2.0	5.8 (5.9)	≤ -1.0	6.1 (6.0)	3.8	3.0
Opus 4.6, with reasoning (high)	1.0	3.5	35	≤ -1.0	29	3.8	13
Gemma 4 31B, without reasoning	0.0 (−0.2)	0.5 (0.2)	2.0 (1.6)	≤ -0.3	1.0 (0.7)	0.8 (0.4)	1.0 (0.6)
Gemma 4 31B, with reasoning	1.0	3.5	35	≤ -1.0	21	4.0	14
Qwen3.5 9B, without reasoning	−0.2 (−0.8)	2.7 (0.4)	5.5 (1.1)	≤ -1.0	1.6 (−0.7)	2.3 (−0.6)	4.1 (1.1)
Qwen3.5 9B, with reasoning	0.9	3.5	> 60 (34)	≤ -1.0	29 (27)	3.9	15 (14)
gpt-oss-20b, without reasoning	0.5 (0.3)	1.8 (1.3)	2.7 (1.7)	≤ -0.5	1.8 (1.5)	1.7 (1.5)	2.5 (1.6)
gpt-oss-20b, with reasoning (low)	0.3	1.0	> 60 (24)	≤ -1.0	1.3 (1.2)	8.8	8.6 (8.4)
Haiku 5.5, with reasoning (medium)	1.1	3.3 (3.5)	28	≤ -1.0	33	3.8 (3.9)	8.5 (8.8)
Haiku 5.5, with reasoning (low)	0.8 (0.7)	3.2 (3.3)	24	≤ -1.0	27 (26)	5.5 (5.6)	8.6 (8.4)
Olmo 3 7B Think, with reasoning	−0.5	0.0	6.2 (6.0)	≤ -1.0	5.9 (5.6)	5.5 (5.3)	15 (13)

coefficient averages over replies, whereas per-response weights show which prediction rule each reply follows. We use the tasks with $|m - \bar{y}| \geq 30$, of which there are 61 at $c = 36$ and 124 at $c = 1$, and we exclude replies without a number. A reply is classified as consistent with a rule when its per-response weight is within 0.05 of that rule's weight. The candidate weights are the Bayesian reference (0.90 for $c = 36$ and 0.20 for $c = 1$); the reference with its two precisions exchanged (0.10 and 0.80); the description counted as one additional observation, $1/(n + 1)$; the description alone, with weight 1; and the demonstrations alone, with weight 0. The one-additional-observation rule coincides with the reference at $c = 1$.

With reasoning, Opus 4.6, GPT-6 Sol, Gemma 4 31B, and Haiku 5.5 follow the reference rule (Section 7.2); without reasoning, 77% of Opus 4.6 replies at $c = 36$ match none of these candidate rules, as do more than half of GPT-6 Sol replies. Olmo 3 7B Think averages demonstrations well but places more weight than the reference on the $c = 1$ and $c = 4$ descriptions. At $n = 1$, it repeats the description value in 57 to 80% of replies, more often at higher stated precision. At $n = 4$, about half of its eligible replies return the description value alone and a third match the reference. Without a description, its root-mean-square error falls from 56 at $n = 1$ to 10.8 at $n = 32$, compared with 10.0 for the reference. Its shortfall in ESS (Section 7.2) therefore lies in combining the description with demonstrations.

Within each cell, Haiku 5.5's replies split by whether they include reasoning output. At $n = 1$ and at either effort, replies with reasoning output assign weights 0.50, 0.80, and 0.97 to the relevant $c = 1$, 4, and 36 descriptions; these are the reference weights to two decimals. Replies without reasoning output assign weight 0.00 to 0.05 and repeat the single demonstration value. Its fitted weight at $n = 1$ mixes the two kinds of reply: 0.44, 0.66, and 0.74 at medium effort. On these three descriptions, 34 to 53% of its replies include reasoning output at low effort, against 72 to 82% at medium effort. Its fitted weights at $n = 1$ are therefore lower at low effort: 0.36, 0.34, and 0.41, nearly the same for all three precisions. This holds even at $c = 1$ and $c = 4$, where its summary ESS is close to the reference's (Table 7.6). With more demonstrations, more replies include reasoning output, and its weight approaches the reference again: at $n = 4$ and low effort, it is 0.17, 0.48, and 0.86, compared with 0.20, 0.50, and 0.90. Whether a reply includes reasoning output is the model's own choice, so this split is observational rather than experimental.

Fixed-weight averaging. Table 7.7 reports the fitted description weights behind Section 7.3, for $c = 36$ under the original warning, together with the weight $p\,36/(36+n)$ that fixed-weight averaging gives in the scalar approximation. GPT-6 Sol, Gemma 4 31B, and Qwen3.5 9B track fixed-weight averaging at the stated 50%. Opus 4.6 and Haiku 5.5 at medium effort also fall below the Bayesian reference weight by $n = 4$. At the stated 50% and $n = 2$ and 4, Haiku 5.5 at low effort comes closest to the reference; at $n = 1$, where its replies often contain no reasoning output, its weight equals that of fixed-weight averaging. gpt-oss-20b assigns nearly the weight of a relevant description regardless of the stated probability. The individual replies show the same pattern. Pooling $n = 1$, 2, and 4 at the stated 50%, the share of replies within 2 of the fixed-weight prediction is 100% for Gemma 4 31B, 97% for Qwen3.5 9B, 89% for GPT-6 Sol, 80% for Opus 4.6, 58% for Haiku 5.5 at medium effort, and 21% for gpt-oss-20b. The share within 2 of the reference prediction is at most 24% for every model. Regressing the replies' normalized description weights on the reference's π_n, as described below, gives slopes of 0.01 for GPT-6 Sol, 0.02 for Gemma 4 31B, 0.06 for Qwen3.5 9B, 0.22 for Haiku 5.5, 0.30 for Opus 4.6, and 0.00 for gpt-oss-20b, where updating as the reference does gives 1.

Stating the baseline prior. The matched design of Section 7.3 uses the same 200 tasks, $c = 36$, stated irrelevance probabilities of 10% and 50%, and $n = 1$, 2, and 4. Each model answers 1,200 prompts with the task's relevant description and 1,200 with an irrelevant one. This subsection concerns the relevant descriptions, which differ from the original runs only in stating the baseline prior. Stating it is needed because the reference's assumption matters under the original warning: with its baseline curve held fixed, its summary ESS at the stated 50% is 5.7 under our uniform prior over 100 to 999, 4.6 under the generating distribution $\mathcal{N}(500, 100^2)$, and 13.4 under a uniform prior over 0 to 2000. Table 7.8 compares the fitted description weights with those from the original runs.

At the stated 50%, the reference and fixed-weight predictions differ by at least 5 on 123 to 155 tasks per cell; these are the tasks on which Section 7.3 reports the share of replies closer to the reference. At the stated 10%, at most 7 tasks per cell separate the two rules by this amount, so the comparison is uninformative there.

The normalized description weight of Section 7.3 is computed from each reply's per-response weight, on tasks with $|m - \bar{y}| \geq 30$. For the scalar reference, it equals π_n

Table 7.7: **Description weights under stated unreliability.** Fitted coefficients on the $c = 36$ description value for the first 200 tasks, with stated irrelevance probability $1 - p$. All model rows use reasoning. "Fixed weights" uses weight $p\,36/(36 + n)$; the reference updates the relevance probability using a uniform baseline prior over 100 to 999. Bootstrap standard errors are at most 0.03 at $n = 4$ in the stated-50% condition and at most 0.06 overall, except for gpt-oss-20b, whose maximum is 0.09. At the stated 50%, several reasoning configurations closely track fixed-weight averaging.

	Stated irrelevance 50%			Stated irrelevance 10%		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Fixed weights	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol	0.49	0.46	0.40	0.90	0.86	0.71
Opus 4.6	0.56	0.43	0.40	0.93	0.86	0.77
Gemma 4 31B	0.49	0.46	0.45	0.89	0.68	0.76
Qwen3.5 9B	0.49	0.46	0.43	0.88	0.84	0.80
Haiku 5.5, medium	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, low	0.49	0.62	0.58	0.42	0.94	0.88
gpt-oss-20b	0.89	0.87	0.82	0.95	0.87	0.90

up to the negligible boundary effects of Section 7.1. For model predictions, the ratio can lie outside $[0, 1]$ and does not identify an internal probability. Here π_n ranges from 0.08 to 0.88 as the demonstrations provide different evidence for relevance. The regression pools $n = 1$, 2, and 4 at the stated 50%, with 269 replies per model, and its 95% intervals resample the 153 tasks. With the prior stated, the intercepts are 0.01 and 0.03, and R^2 is 0.57 and 0.33.

The agreement also holds when the demonstrations count against the description. Among the 23 replies with π_n below 0.5 (mean 0.33), the mean normalized weights are 0.30 and 0.38 for the two models with the baseline prior stated, compared with 0.48 and 0.50 under the original warning. Both models therefore vary their description weights across tasks in approximately the amount predicted by Bayesian model averaging.

Summary ESS uses each model's baseline curve from the main run. At the stated 50%, fixed-weight averaging gives 4.4 and the Bayesian reference gives 5.7. Stating the baseline prior raises ESS from 4.2 to 5.4 for GPT-6 Sol and from 3.8 to 4.2 for Haiku 5.5. At the stated 10%, it changes GPT-6 Sol's ESS from 29.7 to 25.0, and lowers Haiku 5.5's to 15.1, against 27.5 for the reference (Section 7.3). The two models' fitted weights agree within 0.01 in this condition (Table 7.8). Haiku 5.5's lower ESS therefore does not reflect lower average reliance on the description; it comes from a minority of replies far from the reference, some of which repeat a demonstration value. Two further configurations were run on the relevant descriptions with the baseline prior stated. Haiku 5.5 at low effort also updates (slope 0.93), with summary ESS 2.7 at the stated 50%, against 5.5 under the original warning. Opus 4.6 without reasoning does not (slope -0.18), with summary ESS 3.8 under both warnings.

7.5.7 Wording with Reasoning Enabled

Because reasoning brings description weights close to the reference under the original wording, we test whether it does so under the alternative wordings as well. We evaluate the three alternative wordings with Haiku 5.5 at medium effort, for the three relevant descriptions at $n = 1$, 2, and 4 on the first 200 tasks; all 5,400 replies contain a number. With "An earlier estimate put its mean at m, with a standard error of τ", at least 71% of replies in each cell include reasoning output; 99% of these lie within 1 of the reference prediction. Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$", reasoning output is absent from 4 to 36% and 2 to 74% of replies per cell, respectively.

Table 7.8: **Stating the baseline prior changes description weights.** Fitted coefficients on the $c = 36$ description value for the first 200 tasks, as in Table 7.7. "Unstated" denotes the original warning, "the mean is unknown"; "stated" specifies that the mean is otherwise uniform over 100 to 999, as the reference assumes. GPT-6 Sol uses low effort and Haiku 5.5 uses medium effort.

	Stated irrelevance 50%			Stated irrelevance 10%		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Fixed weights	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol, unstated	0.49	0.46	0.40	0.90	0.86	0.71
GPT-6 Sol, stated	0.58	0.59	0.59	0.88	0.83	0.80
Haiku 5.5, unstated	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, stated	0.62	0.60	0.63	0.89	0.84	0.80

Even among replies with reasoning output, 71% and 91% repeat the description value, while 31% and 20% lie within 1 of the reference prediction. At $c = 1$ and $n = 4$, Haiku 5.5's description weights under these two wordings are 0.72 and 0.93, compared with reference weight 0.20.

All wordings provide a value m and a scale τ, but "give or take" and "±" do not identify τ as a prior standard deviation or specify a distribution for the mean. As without reasoning (Section 7.5.4), the comparison therefore varies how explicit the probabilistic specification is, as well as how it is worded. The results show that accurate combination of descriptions and demonstrations under explicit statistical statements need not extend to these less explicit formulations.

7.5.8 Details of the Semantic Task

Task. A message has six binary properties: its sender is outside the organization; it requests a payment; it concerns travel; it requests computer access; it has a deadline within a day; and it contains personal contact information. Each message states every property as present or absent, using one of three phrasings for each value, such as "I'm a contractor from another company." or "I'm on the sales team here." Properties are sampled independently with probability one half, phrasings are sampled uniformly, and sentences appear in random order.

A policy uses exactly two properties and requires each to be present or absent. A

message goes into folder A when both requirements hold, and into B otherwise. There are $\binom{6}{2}2^2 = 60$ such policies. The prompt describes this family and states that the company's policy was chosen uniformly from it. It then provides a policy note, n previously filed messages with their folders, and four new messages. The model is asked for each new message's probability of going into A, as a number between 0 and 1, on four lines from "1: <probability>" to "4: <probability>". Without reasoning, GPT-6 Sol identifies each property correctly in 97–100% of 96 messages; all three misreadings concern one phrasing, "I'm not traveling for this."

Policy notes. A broad note reads, for example, "Policy note: whether it concerns travel is one of the two properties the policy uses," leaving 20 possible policies. A specific note reads "Policy note: to go into A, a message must request a payment. The other property the policy uses is whether it concerns travel," leaving 2.

The unreliable note adds "This note has a 50% chance of being wrong, in which case it says nothing about the policy, which is then any of the 60, equally likely." In the generator, the note is relevant with probability one half and otherwise describes an independently drawn policy. The irrelevant component occurs in 92 of the 200 tasks. An independently generated note can agree with the actual policy by chance. Thus, *relevance* denotes the latent generating component, whereas *compatibility* denotes whether a policy satisfies the note; the two are not interchangeable.

Reference and ESS. For a note asserted to be relevant, the reference takes the policies that are consistent with the note and the filed messages, and predicts the fraction of them that assign the new message to A. For an unreliable note, it averages this prediction with the prediction from policies consistent with the filed messages alone, weighting the two by the posterior relevance probability. The prior relevance probability is 0.5; the posterior probability is obtained using the likelihood of the demonstrations under each component. On 3,000 simulated tasks, the reference's predictive probabilities are calibrated in every condition, and its posterior probability that the unreliable note is relevant matches the frequency of relevance: where it exceeds 0.7, it averages 0.80, and the note is relevant in 80% of cases.

The loss is the squared error between the predicted probability of A and the true folder indicator, averaged over the four query messages. ESS follows Definition 3.1. Each learner's baseline curve is its loss without a note, evaluated at $n = 0$, 2, 4, 6,

8, 12, 16, 20, 24, and 32 and interpolated linearly in n. On simulated tasks, this grid changes the reference ESS by at most 0.6 relative to evaluation at every integer demonstration count.

Runs. We use 200 tasks, each defined by a sampled policy, and 19 conditions: no note at the ten demonstration counts above, and each note at $n = 0$, 2, and 4. This gives 3,800 prompts per configuration. Haiku 5.5 runs at medium effort; GPT-6 Sol runs without reasoning and at low effort. The command-line agents follow Section 7.5.5, with no tools, and no reply calls a tool. Fourteen replies from GPT-6 Sol at low effort contain no probability and are excluded; one Haiku 5.5 reply omits one of its four probabilities. Every Haiku 5.5 reply includes reasoning output, with a median of 3,008 reasoning tokens. For GPT-6 Sol at low effort, 91% of replies include reasoning output, with a median of 142 reasoning tokens.

Results. Table 7.9 reports each note's ESS. With reasoning, both models are within 2 demonstrations of the reference in every condition except GPT-6 Sol's specific note at $n = 4$: its ESS is 26.1, compared with 14.8 for the reference. Its loss with this note, 0.008, is close to the reference's 0.007, but its baseline curve flattens beyond $n = 16$, with loss 0.022 at $n = 20$, compared with 0.004 for the reference. It consequently needs more demonstrations on its own baseline curve to match the note's error reduction.

For the unreliable note, fixed-weight averaging (Section 7.4) averages the two component predictions with the stated 50% weights. The one exception is when the filed messages rule out every policy compatible with the note: it then assigns zero weight to the note-conditioned component. We use query messages on which this predictor and the reference differ by at least 0.05: about 150 at $n = 2$ and 200 at $n = 4$, out of 800. The share of predictions closer to the reference is 90% at $n = 2$ and 99% at $n = 4$ for Haiku 5.5; 77% and 87% for GPT-6 Sol at low effort; and 45% at both counts for GPT-6 Sol without reasoning.

Without reasoning, GPT-6 Sol learns slowly from demonstrations. Its loss without a note is still 0.078 at $n = 32$, compared with 0.000 for the reference at the reported precision. Measured against this weaker baseline, a note can have an ESS as large as the reference's or larger. For example, the specific note has ESS 13.5 at $n = 2$, although the model's loss with that note is 0.130, compared with 0.027 for the reference. This illustrates why learner-specific ESS must be interpreted alongside absolute prediction

error.

Table 7.9: **ESS of policy notes in the semantic task.** Results use 200 tasks and each learner's own baseline curve, measured without a note. "≤ 0" means that the note did not lower loss; "> 28" means that the baseline curve does not reach the note-conditioned loss by $n = 32$. On 6,000 simulated tasks, the reference ESS is 9.6 and 13.2 for the specific note at $n = 2$ and 4, and 2.5 and 2.9 for the unreliable note. Models are compared with the reference on the same 200 tasks. Paired 95% bootstrap intervals over tasks (1,000 replicates) for the specific note at $n = 2$ are: reference [9.7, 12.9], Haiku 5.5 [8.7, 11.5], GPT-6 Sol at low effort [7.8, 12.7], and GPT-6 Sol without reasoning [3.7, 21.9].

Note	n	Reference	Haiku 5.5 medium	GPT-6 Sol low	GPT-6 Sol no reasoning
Broad	0	0.0	0.0	≤ 0	≤ 0
	2	1.4	1.3	1.3	0.7
	4	1.8	1.6	1.7	8.6
Specific	0	3.7	3.6	3.3	3.7
	2	11.4	9.9	9.5	13.5
	4	14.8	14.5	26.1	> 28
Unreliable	0	1.3	1.3	1.6	1.7
	2	3.0	2.6	3.2	1.7
	4	5.0	4.9	5.1	2.7

Chapter 8

Discussion

We asked how natural-language descriptions can be characterized and how well learners use them in practice. For the first question, we treated a description as a prior with a specificity and a reliability, and we measured it by its predictive ESS. A description's value then depends on its specificity, its reliability, and the demonstrations already available. In our Gaussian model, a reliable description can replace many demonstrations by constraining the task across directions. Uncertain relevance imposes a finite ESS ceiling before demonstrations are observed. Demonstrations can then raise a description's ESS, by supplying evidence about its relevance or by constraining the directions that a partial description leaves unspecified.

For the second question, meta-trained Transformers approach the Bayes-optimal ESS, and their fitted description weight follows the reliability of their training data. Most of the pretrained language models that we test need reasoning enabled to use a stated precision as the Bayesian reference does. When descriptions are unreliable and the prompt says so, GPT-6 Sol with reasoning also matches the reference, and for the two models tested in this design, the agreement extends from numerical predictions to semantic policy constraints. When the prompt does not say how the task is distributed if the description is irrelevant, or expresses uncertainty informally, the models must supply what is missing from their pretrained knowledge, and their predictions depart from the reference.

The analysis explains, in a controlled setting, why descriptions sometimes substitute for many examples (Le Scao & Rush, 2021; Honda et al., 2025) and sometimes have limited influence (Webson & Pavlick, 2022; Gupta et al., 2025; Gloaguen et al., 2026). A description can have little influence for three different reasons: it says little about

the task, it may be wrong, or the learner does not use what it says. The Bayes-optimal ESS reflects the first two, and comparing a learner's ESS with it reveals the third. More broadly, because a description enters as a prior, the Bayesian analysis of ICL extends from demonstrations to descriptions.

8.1 Scope and Limitations

The theory assumes Gaussian linear regression, squared error, and a particular mixture model of irrelevance. It also covers one kind of description: when relevant, a description is a single Gaussian around one center. Descriptions can take other shapes; one that names several candidate tasks, for example, is a mixture of Gaussians, as in the task prior of Lin & Lee (2024). Future work could cover other kinds of description with a mixture of Gaussians, which can approximate a wide class of priors (Dalal & Hall, 1983). Learner-specific ESS measures improvement relative to a learner's own baseline curve and should be read alongside absolute error. The language-model tasks are small and controlled; the main numerical comparison uses 200 tasks, and the matched design and the semantic experiment cover two models. Some reasoning comparisons also differ in serving, decoding, or readout settings, and the frontier models are served through command-line agents (Section 7.5.5). Agreement with a Bayesian reference concerns a model's predictions: it does not show that the model computes them as the reference does.

The networks are trained only at $d = 5$ and, for the higher-dimension experiment, at $d = 10$. Because the numerical language-model task is scalar, it has no dimension term and no partial description. The language models we test are post-trained chat models, and the wording test with reasoning enabled covers only one of them, Haiku 5.5. The warnings of the earlier design leave the baseline prior unspecified, so the Bayesian reference for them rests on an assumed prior (Section 7.3). Models without reasoning are evaluated through either the distribution readout or the written readout, so comparisons across these models mix differences between models with differences between readouts. How a model infers the precision of a description written in natural language remains an open question. Descriptions that leave their uncertainty unstated, more structured tasks, and richer or heavy-tailed priors (Dalal & Hall, 1983; O'Hagan & Pericchi, 2012) would test how far these findings extend.

8.2 Future Work

We outline four directions for future work. The first two follow from the limitations and findings above, the third from the descriptions studied here, and the fourth from the networks of Chapter 6. None of them is established in this thesis.

Partial descriptions and the dimension term in language models. A precise partial description can become more valuable as demonstrations resolve what it leaves unspecified (Proposition 4.2, Table 4.1), whereas the ESS of a reliable complete description falls toward $c - a_0^{-1}$ (Proposition 4.1(iii)). The networks of Chapter 6 capture this increase with $k = 3$ of $d = 5$ directions described, and their ESS at $c = 10$ is larger at $d = 10$ than at $d = 5$. The numerical language-model task, however, has a single unknown mean. A task with several unknown quantities, of which the description states only some, would test whether a language model's ESS has a dimension term, and whether a partial description becomes more valuable to it as demonstrations accumulate.

Updating reliability. With reasoning, GPT-6 Sol and Haiku 5.5 revise a stated chance of irrelevance in light of the demonstrations once the prompt says how the mean is distributed when the description is irrelevant (Section 7.3). In the matched design, where descriptions are in fact irrelevant at the stated rate, GPT-6 Sol's ESS is within 0.2 of the reference at both stated chances. Haiku 5.5 falls short: at a 10% chance, its ESS is 3.2 against 5.0, and only 60% of its replies to an irrelevant description are within 2 of the reference. On the relevant descriptions alone, its ESS is 15.1, against 27.5 for the reference. Its fitted weights there agree with GPT-6 Sol's within 0.01, so the shortfall does not come from lower average reliance on the description: a minority of its replies fall far from the reference, some repeating a demonstration value (Section 7.5.6). The matched design covers two models, one precision ($c = 36$), and at most four demonstrations. Two questions remain open. Do the other reasoning models revise the chance once the prompt states the irrelevant case? And why do some of Haiku 5.5's replies depart so far from the reference?

Structural descriptions. Apart from the semantic task, the descriptions studied here state where the task lies, through a center m with a stated precision. Many

instructions instead state what kind of task it is, for example that only three features matter, or that the answer is monotone in an input. Such a description narrows the hypothesis class rather than pointing at one task. Our analysis already covers one case of this kind. A description that confines w to a known s-dimensional subspace states the other $k = d - s$ directions exactly, so it is a partial description of those directions in the limit $c \to \infty$. A heuristic calculation suggests that such a description keeps its value as demonstrations accumulate. Under a flat baseline prior, it would lower the Bayes-optimal error from about d/n to about s/n, so its ESS would be about $n(d/s - 1)$, which grows with the number of demonstrations. If a structural description is irrelevant, however, the Bayesian predictor that assumes relevance would keep a bias that no number of demonstrations removes. Our analysis does not cover structures that fix no direction, such as sparsity with unknown relevant features, or monotonicity. The definition of ESS and the network and language-model tests apply to such structures as well. The semantic task of Section 7.4 gives a first test in a language model. The task's broad note names a property that the policy uses without stating the value that the property must take. With reasoning, the broad note is worth 1.3 demonstrations to Haiku 5.5 and to GPT-6 Sol at $n = 2$, against 1.4 for the reference (Table 7.9).

Where a network stores its description weight. The meta-trained networks take their fitted description weight from the reliability of their training data (Chapter 6). A natural next step is to locate the components of a network that set this weight, and to test by intervention whether changing them changes the weight without retraining. Such a study would connect the behavioral measure of this thesis to the mechanisms by which networks learn in context.

Bibliography

Abdin, M., Aneja, J., Behl, H., et al. Phi-4 technical report. Technical report, Microsoft, 2024. arXiv:2412.08905.

Akata, E., Voudouris, K., Fortuin, V., and Schulz, E. In-context function learning in large language models. In *AISTATS*, 2026.

Akyürek, E., Schuurmans, D., Andreas, J., Ma, T., and Zhou, D. What learning algorithm is in-context learning? Investigations with linear models. In *ICLR*, 2023.

Anthropic. System card: Claude Haiku 5.5. https://www.anthropic.com/claude-haiku-5-5-system-card, 2026a.

Anthropic. System card: Claude Opus 4.6. https://anthropic.com/claude-opus-4-6-system-card, 2026b.

Arora, A., Jurafsky, D., Potts, C., and Goodman, N. D. Bayesian scaling laws for in-context learning. In *COLM*, 2025.

Brown, T. B. et al. Language models are few-shot learners. In *NeurIPS*, 2020.

Cook, J., Sapora, S., Ahmadian, A., Khan, A., Rocktäschel, T., Foerster, J., and Ruis, L. Programming by backprop: an instruction is worth 100 examples when finetuning LLMs. In *ICLR*, 2026.

Cover, T. M. and Thomas, J. A. *Elements of Information Theory*. Wiley, second edition, 2006.

Dalal, S. R. and Hall, W. J. Approximating priors by mixtures of natural conjugate priors. *Journal of the Royal Statistical Society: Series B (Methodological)*, 45(2): 278–286, 1983.

Evans, M. and Moshonov, H. Checking for prior-data conflict. *Bayesian Analysis*, 1 (4):893–914, 2006.

Falck, F., Wang, Z., and Holmes, C. Is in-context learning in large language models Bayesian? A martingale perspective. In *ICML*, 2024.

Filbry, L., Melo, L. C., and Gal, Y. From prompts to posteriors: Can LLM agents perform Bayesian inference in context? *arXiv:2610.29331*, 2026.

Garg, S., Tsipras, D., Liang, P., and Valiant, G. What can Transformers learn in-context? A case study of simple function classes. In *NeurIPS*, 2022.

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. *Bayesian Data Analysis*. CRC Press, third edition, 2013.

Gemma Team. Gemma 4 technical report. *arXiv:2607.02770*, 2026.

Genewein, T., Wenliang, L. K., Grau-Moya, J., Ruoss, A., Orseau, L., and Hutter, M. Understanding prompt tuning and in-context learning via meta-learning. In *NeurIPS*, 2025.

Gloaguen, T., Mündler-Sasahara, N., Müller, M. N., Raychev, V., and Vechev, M. Evaluating AGENTS.md: are repository-level context files helpful for coding agents? *arXiv:2602.11988*, 2026.

Goodman, N. D., Tenenbaum, J. B., Feldman, J., and Griffiths, T. L. A rational analysis of rule-based concept learning. *Cognitive Science*, 32(1):108–154, 2008.

Gruver, N., Finzi, M., Qiu, S., and Wilson, A. G. Large language models are zero-shot time series forecasters. In *NeurIPS*, 2023.

Gupta, R., Corona, R., Ge, J., Wang, E., Klein, D., Darrell, T., and Chan, D. M. Enough coin flips can make LLMs act Bayesian. In *ACL*, 2025.

He, Z., Qiu, T., Shirado, H., and Sap, M. Martingale Score: An unsupervised metric for Bayesian rationality in LLM reasoning. In *NeurIPS*, 2025.

Hoeting, J. A., Madigan, D., Raftery, A. E., and Volinsky, C. T. Bayesian model averaging: A tutorial. *Statistical Science*, 14(4):382–417, 1999. doi: 10.1214/ss/1009212519.

Honda, U., Murakami, S., and Zhang, P. Distilling many-shot in-context learning into a cheat sheet. In *Findings of EMNLP*, 2025.

Huang, R. and Ge, R. Task descriptors help Transformers learn linear models in-context. In *ICLR*, 2025.

Kwa, T., West, B., Becker, J., Deng, A., Garcia, K., Hasin, M., Jawhar, S., Kinniment, M., Rush, N., Von Arx, S., et al. Measuring AI ability to complete long software tasks. *arXiv:2503.14499*, 2025.

Le Scao, T. and Rush, A. M. How many data points is a prompt worth? In *NAACL*, 2021.

Lin, Z. and Lee, K. Dual operating modes of in-context learning. In *ICML*, 2024.

Lin, Z., Bharti, S. K., and Lee, K. In-context learning with hypothesis-class guidance. *arXiv:2502.19787*, 2025.

Liu, T. J. B., Boullé, N., Sarfati, R., and Earls, C. J. Density estimation with LLMs: a geometric investigation of in-context learning trajectories. In *ICLR*, 2025.

Mao, Y., He, J., and Chen, C. From prompts to templates: A systematic prompt template analysis for real-world LLMapps. In *FSE Industry Track*, 2025. arXiv:2504.02052.

Min, S., Lyu, X., Holtzman, A., Artetxe, M., Lewis, M., Hajishirzi, H., and Zettlemoyer, L. Rethinking the role of demonstrations: What makes in-context learning work? In *EMNLP*, 2022.

MiniCPM Team. MiniCPM5-2B. https://huggingface.co/openbmb/MiniCPM5-2B, 2026. Model card; the series is described in *MiniCPM4: Ultra-Efficient LLMs on End Devices*, arXiv:2506.07900.

Morita, S., Thall, P. F., and Müller, P. Determining the effective sample size of a parametric prior. *Biometrics*, 64(2):595–602, 2008.

Muirhead, R. J. *Aspects of Multivariate Statistical Theory*. Wiley, 1982.

Müller, S., Hollmann, N., Pineda Arango, S., Grabocka, J., and Hutter, F. Transformers can do Bayesian inference. In *ICLR*, 2022.

Neuenschwander, B., Weber, S., Schmidli, H., and O'Hagan, A. Predictively consistent prior effective sample sizes. *Biometrics*, 76(2):578–587, 2020.

O'Hagan, A. and Pericchi, L. Bayesian heavy-tailed models and conflict resolution: A review. *Brazilian Journal of Probability and Statistics*, 26(4):372–401, 2012.

OpenAI. gpt-oss-120b & gpt-oss-20b model card. Technical report, OpenAI, 2025. arXiv:2508.10925.

OpenAI. GPT-6 Sol and GPT-6 Luna: October 2026 update. https://deployment safety.openai.com/gpt-6-october, 2026.

Panwar, M., Ahuja, K., and Goyal, N. In-context learning through the Bayesian prism. In *ICLR*, 2024.

Patsakis, C., Argyropoulos, V., and Alepis, E. Configuration, not conscience: A large-scale empirical study of LLM system prompts. In *ICTAI*, 2026. arXiv:2609.31575.

Qiu, L., Sha, F., Allen, K., Kim, Y., Linzen, T., and van Steenkiste, S. Bayesian teaching enables probabilistic reasoning in large language models. *Nature Communications*, 17:1238, 2026.

Qwen Team. Qwen3.5-9B. https://huggingface.co/Qwen/Qwen3.5-9B, 2026. Model card.

Raventós, A., Paul, M., Chen, F., and Ganguli, S. Pretraining task diversity and the emergence of non-Bayesian in-context learning for regression. In *NeurIPS*, 2023.

Reimherr, M., Meng, X.-L., and Nicolae, D. L. Prior sample size extensions for assessing prior impact and prior-likelihood discordance. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83(3):413–437, 2021.

Requeima, J., Bronskill, J., Choi, D., Turner, R. E., and Duvenaud, D. LLM processes: numerical predictive distributions conditioned on natural language. In *NeurIPS*, 2024.

Reznik, Y. A. The optimal discounting parameter of the power prior under predictive log-loss. *arXiv:2608.12159*, 2026.

Schmidli, H., Gsteiger, S., Roychoudhury, S., O'Hagan, A., Spiegelhalter, D., and Neuenschwander, B. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics*, 70(4):1023–1032, 2014.

Selby, D., Iwashita, Y., Spriestersbach, K., Saad, M., Bappert, D., Warrier, A., Mukherjee, S., Kise, K., and Vollmer, S. Had enough of experts? Quantitative knowledge retrieval from large language models. *Stat*, 14:e70054, 2025.

Team Olmo. Olmo 3. *arXiv preprint arXiv:2512.13961*, 2025.

Team OLMo, Walsh, P., Soldaini, L., Groeneveld, D., Lo, K., et al. 2 OLMo 2 Furious. In *COLM*, 2025. arXiv:2501.00656.

Tong, X., Zeng, Y., and Zhang, J. Demonstrations, CoT, and prompting: a theoretical analysis of ICL. *arXiv:2603.19611*, 2026.

von Oswald, J., Niklasson, E., Randazzo, E., Sacramento, J., Mordvintsev, A., Zhmoginov, A., and Vladymyrov, M. Transformers learn in-context by gradient descent. In *ICML*, 2023.

von Rosen, D. Moments for the inverted Wishart distribution. *Scandinavian Journal of Statistics*, 15(2):97–109, 1988.

Webson, A. and Pavlick, E. Do prompt-based models really understand the meaning of their prompts? In *NAACL*, 2022.

Wei, J., Wei, J., Tay, Y., Tran, D., Webson, A., Lu, Y., Chen, X., Liu, H., Huang, D., Zhou, D., and Ma, T. Larger language models do in-context learning differently. *arXiv:2303.03846*, 2023.

Weller, O., Lourie, N., Gardner, M., and Peters, M. E. Learning from task descriptions. In *EMNLP*, pp. 1361–1375, 2020. doi: 10.18653/v1/2020.emnlp-main.105.

Wenliang, L. K., Ruoss, A., Grau-Moya, J., Hutter, M., and Genewein, T. Why is prompting hard? Understanding prompts on binary sequence predictors. In *AISTATS*, 2026. PMLR 300:2134–2142.

Wiesenfarth, M. and Calderazzo, S. Quantification of prior impact in terms of effective current sample size. *Biometrics*, 76(1):326–336, 2020.

Xie, S. M., Raghunathan, A., Liang, P., and Ma, T. An explanation of in-context learning as implicit Bayesian inference. In *ICLR*, 2022.

Zhu, Q., Oermann, E. K., and Cho, K. Multi-task Bayesian in-context learning. In *ICML*, 2026. arXiv:2606.20538.

Appendix A

Language-Model Results at Every Number of Demonstrations

This appendix reports the full run of Chapter 7, without reasoning, at every evaluated demonstration count. Tables A.1 and A.2 report ESS for the distribution-readout and written-readout models, and Table A.3 reports description weights for the distribution-readout models. Tables A.1 and A.3 each continue onto a second page.

Table A.1: **ESS by description condition and demonstration count: distribution readout.** Results cover the prompt-based reference and the three distribution-readout models in the full run without reasoning. "Number" is the numeric response rate averaged over n. "10% wrong" and "50% wrong" denote stated irrelevance probabilities. The censoring marker $\leq$ marks error above the first baseline point; $> 64 - n$ means that the description-conditioned error lies below the entire baseline curve through 64 demonstrations.

Predictor	Description	number	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
Reference	$\sigma = 60$, $c = 1$, relevant		1.2	1.3	1.3	1.3	1.2	1.5	> 0
	$\sigma = 60$, $c = 1$, irrelevant		≤ 0.0	-0.5	-0.8	-1.2	-1.7	-1.5	-1.7
	$\sigma = 60$, $c = 4$, relevant		4.5	4.4	3.9	4.6	4.6	5.2	> 0
	$\sigma = 60$, $c = 4$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-20	-32
	$\sigma = 60$, $c = 36$, relevant		36	36	37	38	39	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	≤ -7.0	≤ -15	≤ -31	-62
	$\sigma = 30$, $c = 1$, relevant		1.0	1.2	1.4	1.6	1.4	1.5	> 0
	$\sigma = 30$, $c = 4$, relevant		4.5	4.7	5.0	5.1	4.9	5.6	> 0
	$\sigma = 30$, $c = 36$, relevant		42	43	43	43	42	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong		3.8	3.9	3.6	4.4	4.5	5.1	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong		≤ 0.0	-0.7	-1.2	-1.6	-2.2	-1.9	-2.2
	$\sigma = 60$, $c = 36$, relevant, 10% wrong		13	22	26	35	38	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong		≤ 0.0	-0.9	-1.7	-2.8	-5.4	-6.9	-10
	$\sigma = 60$, $c = 4$, relevant, 50% wrong		1.9	2.2	2.9	3.7	4.1	4.5	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong		≤ 0.0	-0.2	-0.4	-0.5	-0.8	-0.6	-0.7
	$\sigma = 60$, $c = 36$, relevant, 50% wrong		3.1	5.8	12	21	31	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong		≤ 0.0	-0.3	-0.7	-1.2	-2.4	-2.8	-4.5
gpt-oss-20b	$\sigma = 60$, $c = 1$, relevant	0.98	1.0	0.9	-0.1	-1.5	-9.2	-25	-57
	$\sigma = 60$, $c = 1$, irrelevant	0.98	≤ 0.0	-0.3	-1.5	-3.6	-10	-25	-59
	$\sigma = 60$, $c = 4$, relevant	0.98	2.3	1.7	2.0	2.9	-7.5	-24	-48
	$\sigma = 60$, $c = 4$, irrelevant	0.97	≤ 0.0	-0.3	-1.4	-3.7	-11	-26	-59
	$\sigma = 60$, $c = 36$, relevant	0.98	2.8	2.3	3.8	17	3.7	-12	-17
	$\sigma = 60$, $c = 36$, irrelevant	0.98	≤ 0.0	-0.5	-1.5	-3.7	-11	-26	-60
	$\sigma = 30$, $c = 1$, relevant	0.99	0.8	0.4	-0.3	-1.8	-9.9	-26	-59
	$\sigma = 30$, $c = 4$, relevant	0.98	1.2	1.3	1.9	3.4	-8.1	-24	-44
	$\sigma = 30$, $c = 36$, relevant	0.99	1.6	2.3	4.7	31	3.0	-13	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	2.2	1.8	0.0	-0.2	-8.1	-24	-52
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.6	-1.1	-2.2	-9.6	-25	-58
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.97	2.4	2.4	0.5	2.8	-3.6	-20	-33
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.7	-1.1	-2.3	-9.7	-25	-59
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.95	2.3	1.7	0.1	-0.5	-8.3	-24	-53
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.94	≤ 0.0	-0.5	-1.1	-2.3	-9.7	-25	-58
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.95	2.8	2.3	0.7	1.2	-4.5	-21	-37
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.95	≤ 0.0	-0.6	-1.1	-2.3	-9.7	-25	-59
Phi-4	$\sigma = 60$, $c = 1$, relevant	0.87	0.8	0.7	-0.2	-0.4	-1.0	-3.1	-35
	$\sigma = 60$, $c = 1$, irrelevant	0.86	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-6.1	-8.8	-40
	$\sigma = 60$, $c = 4$, relevant	0.85	5.8	4.1	2.4	2.1	0.7	-4.6	-33
	$\sigma = 60$, $c = 4$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.7	-6.2	-13	-42
	$\sigma = 60$, $c = 36$, relevant	0.86	15	7.6	4.7	5.6	8.5	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.5	-6.4	-12	-42
	$\sigma = 30$, $c = 1$, relevant	0.87	1.1	0.8	-0.3	-0.3	-1.0	-1.8	-15
	$\sigma = 30$, $c = 4$, relevant	0.86	4.8	3.5	2.0	1.6	0.6	-1.9	-2.2
	$\sigma = 30$, $c = 36$, relevant	0.88	9.7	6.0	3.8	4.4	8.4	17	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.87	6.0	5.5	5.3	8.3	15	> 32	> 0

	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	$0.85 \leq$	$0.0 \leq$	-1.0	-2.1	-1.2	2.9	-3.2	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.87	10	8.7	8.2	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	$0.86 \leq$	$0.0 \leq$	-1.0	-2.4	-1.7	1.3	-4.7	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.87	6.0	6.4	7.7	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	$0.85 \leq$	$0.0 \leq$	-1.0	-1.3	1.5	16	> 32	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.88	9.2	9.7	11	> 56	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	$0.86 \leq$	$0.0 \leq$	-1.0	-1.8	0.6	13	> 32	> 0
OLMo 2 32B	$\sigma = 60$, $c = 1$, relevant	0.90	1.9	1.8	2.0	2.3	5.8	> 32	> 0
	$\sigma = 60$, $c = 1$, irrelevant	$0.89 \leq$	0.0	-0.5	-0.2	-0.2	1.7	25	> 0
	$\sigma = 60$, $c = 4$, relevant	0.91	5.8	3.4	3.4	4.5	9.3	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant	$0.90 \leq$	0.0	-0.7	-0.3	-0.5	0.3	17	> 0
	$\sigma = 60$, $c = 36$, relevant	0.91	11	4.6	5.3	6.8	15	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	$0.90 \leq$	$0.0 \leq$	-1.0	-0.6	-0.5	-0.2	15	> 0
	$\sigma = 30$, $c = 1$, relevant	0.92	1.8	1.9	1.7	1.7	4.4	24	> 0
	$\sigma = 30$, $c = 4$, relevant	0.92	4.8	3.3	3.1	4.0	9.5	> 32	> 0
	$\sigma = 30$, $c = 36$, relevant	0.93	7.0	4.4	4.0	5.9	14	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	6.0	4.6	3.8	4.5	5.8	23	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	$0.96 \leq$	0.0	-0.9	0.0	0.0	-0.4	4.1	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.96	12	5.9	6.0	6.3	8.9	31	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	$0.96 \leq$	$0.0 \leq$	-1.0	-0.3	-0.1	-0.8	0.3	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.93	5.3	4.2	3.5	4.0	5.4	27	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	$0.92 \leq$	0.0	-0.5	0.1	0.1	0.0	11	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.93	7.8	5.4	4.9	5.6	7.9	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	$0.92 \leq$	0.0	-0.8	-0.1	0.0	-0.3	7.8	> 0

Table A.2: **ESS by description condition and demonstration count: written readout.** Results cover the three written-readout models in the full run without reasoning. Replies without a qualifying number are excluded. Columns and reference values follow Table A.1.

Predictor	Description	number	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Qwen3.5 9B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.2	0.3	−0.9	−2.8	−9.3	−24	−51
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.7	−4.8	−9.7	−23	−49
	$\sigma=60$, $c=4$, relevant	1.00	3.9	2.7	1.3	−0.7	−6.2	−18	−44
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	≤ -1.0	−1.7	−3.8	−9.4	−24	−49
	$\sigma=60$, $c=36$, relevant	1.00	6.5	4.0	2.8	6.9	17	>32	>0
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.8	−5.1	−10	−24	−51
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	0.2	−0.7	−2.5	−8.7	−22	−49
	$\sigma=30$, $c=4$, relevant	1.00	3.4	2.7	1.5	−0.3	−4.5	−14	−40
	$\sigma=30$, $c=36$, relevant	1.00	4.7	4.3	3.0	9.2	14	>32	>0
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	3.8	−0.4	−0.7	−1.8	−8.5	−21	−47
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−4.9	−10	−24	−50
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	5.3	0.4	0.6	−0.3	−3.3	−4.2	−6.9
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−5.2	−10	−25	−53
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	4.5	−0.1	−0.4	−1.9	−9.0	−23	−50
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−6.2	−12	−26	−54
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	7.3	0.8	1.0	−0.1	−4.0	−4.2	−12
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−7.0	−13	−27	−56
Gemma 4 12B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.0	0.0	−0.1	−0.6	−3.2	−5.4	−21
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	−0.4	−1.3	−1.3	−4.4	−6.8	−24
	$\sigma=60$, $c=4$, relevant	1.00	0.3	0.5	1.4	1.1	−0.6	−2.7	−13
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	−0.4	−1.1	−1.2	−4.1	−6.9	−23
	$\sigma=60$, $c=36$, relevant	1.00	1.8	1.2	2.7	3.6	3.6	4.9	>0
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	−0.8	−1.3	−1.3	−4.9	−7.9	−26
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	−0.2	−0.1	−0.5	−3.2	−7.9	−27
	$\sigma=30$, $c=4$, relevant	1.00	0.7	0.8	2.1	1.5	−0.2	−4.6	−18
	$\sigma=30$, $c=36$, relevant	1.00	4.0	1.7	3.4	3.7	2.4	−1.8	−8.9
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	0.1	0.7	1.4	0.4	−1.0	−3.1	−7.3
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.4	−0.5	−2.2	−5.4	−13
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	0.6	1.3	2.3	1.6	0.7	−0.6	−1.3
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.5	−0.5	−2.5	−5.8	−15
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	1.0	0.6	1.7	1.0	−0.1	−1.7	−4.5
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.4	−0.4	−1.3	−4.1	−9.6
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	5.1	1.0	2.7	2.3	2.4	1.1	>0
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.6	−0.4	−1.9	−4.8	−13
Gemma 4 31B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.1	0.0	−0.1	−0.5	−1.4	−5.1	−13
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	−0.1	−0.1	−0.5	−1.0	−4.5	−11
	$\sigma=60$, $c=4$, relevant	1.00	0.8	0.1	0.2	−0.5	−1.0	−4.3	−12
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	−0.2	−0.2	−0.4	−1.0	−4.4	−10
	$\sigma=60$, $c=36$, relevant	1.00	2.6	0.6	1.4	0.9	2.9	3.4	−2.9
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−0.8	−0.6	−1.1	−2.7	−4.8
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	0.0	−0.1	−0.7	−1.2	−3.0	−8.1
	$\sigma=30$, $c=4$, relevant	1.00	0.5	0.0	0.1	−0.5	−1.3	−3.4	−8.3
	$\sigma=30$, $c=36$, relevant	1.00	4.3	1.5	2.1	2.3	3.3	3.3	−0.4
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	0.1	0.0	0.0	−0.7	−1.9	−5.0	−15
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.2	−4.1	−9.5
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	1.0	0.0	0.6	−0.3	−1.4	−4.8	−12
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−0.9	−3.0	−6.2
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	0.2	0.0	0.0	−0.7	−1.7	−4.6	−17
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−1.2	−3.6	−12
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	1.0	0.0	0.5	−0.3	−1.5	−4.5	−15
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.0	−2.9	−8.5

Table A.3: **Description weights by demonstration count: distribution readout.** Results cover the prompt-based reference and the three distribution-readout models in the full run without reasoning. "Number" is defined as in Table A.1; cells are italicized when the numeric response rate is below three quarters at the corresponding n.

Predictor	Description	number	$n=0$	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Reference	$\sigma=60$, $c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60$, $c=1$, irrelevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60$, $c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60$, $c=4$, irrelevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60$, $c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60$, $c=36$, irrelevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=30$, $c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=30$, $c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=30$, $c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60$, $c=4$, relevant, 10% wrong		0.90	0.71	0.59	0.45	0.30	0.19	0.10	0.05
	$\sigma=60$, $c=4$, irrelevant, 10% wrong		0.90	0.27	0.14	0.08	0.04	0.02	0.01	0.00
	$\sigma=60$, $c=36$, relevant, 10% wrong		0.90	0.84	0.84	0.82	0.77	0.65	0.51	0.34
	$\sigma=60$, $c=36$, irrelevant, 10% wrong		0.90	0.26	0.14	0.07	0.03	0.01	0.00	0.00
	$\sigma=60$, $c=4$, relevant, 50% wrong		0.50	0.46	0.39	0.32	0.22	0.14	0.08	0.04
	$\sigma=60$, $c=4$, irrelevant, 50% wrong		0.50	0.12	0.07	0.03	0.02	0.01	0.00	0.00
	$\sigma=60$, $c=36$, relevant, 50% wrong		0.50	0.55	0.58	0.60	0.59	0.53	0.42	0.29
	$\sigma=60$, $c=36$, irrelevant, 50% wrong		0.50	0.12	0.06	0.03	0.01	0.00	0.00	0.00
gpt-oss-20b	$\sigma=60$, $c=1$, relevant	0.98	0.94	0.53	0.29	0.23	0.16	0.12	0.12	0.23
	$\sigma=60$, $c=1$, irrelevant	0.98	0.93	0.59	0.25	0.18	0.09	0.05	0.04	0.10
	$\sigma=60$, $c=4$, relevant	0.98	0.92	0.48	0.32	0.26	0.21	0.19	0.18	0.31
	$\sigma=60$, $c=4$, irrelevant	0.97	0.92	0.57	0.25	0.16	0.08	0.04	0.04	0.09
	$\sigma=60$, $c=36$, relevant	0.98	0.94	0.49	0.35	0.30	0.25	0.23	0.22	0.35
	$\sigma=60$, $c=36$, irrelevant	0.98	0.95	0.60	0.25	0.15	0.07	0.04	0.04	0.09
	$\sigma=30$, $c=1$, relevant	0.99	0.98	0.43	0.23	0.17	0.12	0.09	0.11	0.21
	$\sigma=30$, $c=4$, relevant	0.98	0.98	0.38	0.28	0.23	0.19	0.18	0.19	0.31
	$\sigma=30$, $c=36$, relevant	0.99	0.99	0.37	0.32	0.27	0.23	0.22	0.24	0.37
	$\sigma=60$, $c=4$, relevant, 10% wrong	0.96	0.91	0.45	0.33	0.15	0.13	0.12	0.13	0.22
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	0.96	0.91	0.55	0.34	0.11	0.05	0.02	0.02	0.04
	$\sigma=60$, $c=36$, relevant, 10% wrong	0.97	0.92	0.44	0.34	0.17	0.15	0.15	0.16	0.25
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	0.96	0.92	0.53	0.34	0.10	0.04	0.02	0.02	0.04
	$\sigma=60$, $c=4$, relevant, 50% wrong	0.95	0.79	0.46	0.32	0.16	0.12	0.12	0.12	0.20
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	0.94	0.79	0.53	0.32	0.11	0.04	0.02	0.01	0.04
	$\sigma=60$, $c=36$, relevant, 50% wrong	0.95	0.83	0.46	0.32	0.17	0.14	0.15	0.15	0.23
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	0.95	0.83	0.53	0.31	0.11	0.04	0.02	0.02	0.04
Phi-4	$\sigma=60$, $c=1$, relevant	0.87	*0.97*	0.82	0.62	0.43	0.28	0.18	0.13	0.12
	$\sigma=60$, $c=1$, irrelevant	0.86	*0.97*	0.83	0.64	0.42	0.22	0.11	0.06	0.05
	$\sigma=60$, $c=4$, relevant	0.85	*0.98*	0.76	0.55	0.37	0.26	0.19	0.15	0.16
	$\sigma=60$, $c=4$, irrelevant	0.84	*0.98*	0.79	0.56	0.36	0.18	0.09	0.06	0.05
	$\sigma=60$, $c=36$, relevant	0.86	*0.99*	0.74	0.53	0.38	0.29	0.23	0.20	0.20
	$\sigma=60$, $c=36$, irrelevant	0.84	*0.99*	0.78	0.53	0.33	0.16	0.09	0.06	0.04
	$\sigma=30$, $c=1$, relevant	0.87	*0.99*	0.77	0.58	0.38	0.22	0.13	0.09	0.09
	$\sigma=30$, $c=4$, relevant	0.86	*1.00*	0.69	0.51	0.34	0.23	0.16	0.13	0.13
	$\sigma=30$, $c=36$, relevant	0.88	*1.00*	0.65	0.48	0.32	0.23	0.18	0.17	0.16
	$\sigma=60$, $c=4$, relevant, 10% wrong	0.87	*0.94*	0.62	0.46	0.30	0.21	0.15	0.13	0.11
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	0.85	*0.95*	0.62	0.44	0.26	0.13	0.06	0.04	0.03
	$\sigma=60$, $c=36$, relevant, 10% wrong	0.87	*0.95*	0.61	0.46	0.33	0.25	0.20	0.18	0.16
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	0.86	*0.96*	0.61	0.43	0.26	0.13	0.06	0.04	0.03
	$\sigma=60$, $c=4$, relevant, 50% wrong	0.87	*0.92*	0.57	0.43	0.29	0.20	0.15	0.13	0.11
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	0.85	*0.92*	0.57	0.42	0.24	0.12	0.06	0.04	0.03
	$\sigma=60$, $c=36$, relevant, 50% wrong	0.88	*0.93*	0.56	0.44	0.33	0.24	0.21	0.19	0.17

	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.86	*0.94*	0.57	0.41	0.24	0.12	0.06	0.04	0.03
OLMo 2 32B	$\sigma = 60$, $c = 1$, relevant	0.90	*0.99*	0.72	0.33	0.25	0.16	0.09	0.05	0.05
	$\sigma = 60$, $c = 1$, irrelevant	0.89	*0.99*	0.77	0.29	0.16	0.09	0.05	0.02	0.02
	$\sigma = 60$, $c = 4$, relevant	0.91	*1.00*	0.70	0.37	0.30	0.21	0.14	0.08	0.07
	$\sigma = 60$, $c = 4$, irrelevant	0.90	*1.00*	0.75	0.29	0.15	0.08	0.04	0.02	0.01
	$\sigma = 60$, $c = 36$, relevant	0.91	*1.00*	0.70	0.40	0.34	0.26	0.19	0.12	0.10
	$\sigma = 60$, $c = 36$, irrelevant	0.90	*1.00*	0.76	0.29	0.17	0.08	0.04	0.02	0.02
	$\sigma = 30$, $c = 1$, relevant	0.92	*0.99*	0.69	0.32	0.23	0.14	0.09	0.04	0.04
	$\sigma = 30$, $c = 4$, relevant	0.92	*1.00*	0.65	0.34	0.27	0.19	0.13	0.06	0.06
	$\sigma = 30$, $c = 36$, relevant	0.93	*1.00*	0.64	0.36	0.32	0.23	0.16	0.08	0.09
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	1.01	0.72	0.42	0.29	0.18	0.11	0.09	0.05
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	1.01	0.81	0.32	0.14	0.06	0.03	0.02	0.01
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.96	1.01	0.71	0.44	0.33	0.23	0.15	0.13	0.07
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	1.01	0.82	0.33	0.15	0.07	0.03	0.02	0.01
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.93	1.02	0.65	0.39	0.27	0.16	0.09	0.07	0.04
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.92	1.01	0.71	0.29	0.13	0.06	0.02	0.01	0.01
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.93	1.02	0.64	0.42	0.31	0.21	0.13	0.11	0.06
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.92	1.02	0.72	0.30	0.14	0.06	0.03	0.02	0.01